

STRUCTURAL MODULE 1

STRUCTURAL BASICS OF MASS TIMBER

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STRUCTURAL MODULE 1: Structural Basics of Mass Timber

Structural engineering plays a significant role in the design, performance, economy, and material efficiency of mass timber buildings. This module introduces the foundational principles of mass timber structural engineering, from the behavior of wood as a material to the performance of mass timber structural components, both individually and as part of larger systems. Together, these topics establish the technical foundation for understanding structural capacity, performance criteria, and project constraints before further engaging with design decisions in Structural Module 2: Structural Design with Mass Timber.

Dr. Gregory R. Kingsley is a founding partner and now Director of Research and Development at KL&A Engineers and Builders in Golden, Colorado. His passion is for innovative wood structures and the use of local materials and mass timber as a catalyst for forest health and resilience. Greg has led efforts to advance the acceptance of tall wood buildings in the Denver area, and has spearheaded the first use of Colorado sourced wood in mass timber buildings.

Laura Britton is an architect, author, and educator with a focus on environmentally- and socially-conscious design. As a Senior Associate at Shigeru Ban Architects, she has led and contributed to diverse projects ranging from complex mass timber structures to humanitarian relief efforts. Britton is the Co-Author and Editor of Shigeru Ban: Timber in Architecture, a book tracing the evolution of 45 wood projects from concept through construction. She is an Adjunct Assistant Professor at The Cooper Union and Co-Founder of Mass Timber Pedagogy.

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Wood Structural Basics

WOOD IS HIGHLY VARIED

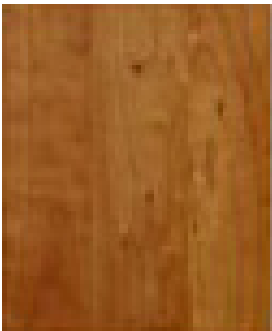
Wood is a natural material with inherent variation, both within a single member and among species.

Unlike concrete and steel, wood does not exhibit uniform material and structural properties. Strength, stiffness, and dimensional stability vary by direction, species, growth conditions, and even within a single structural member. These are defining characteristics of a natural material.

The range of structural wood includes softwoods and hardwoods of many species, each with distinct structural (and aesthetic) characteristics. Structural engineering involves understanding and strategically leveraging this variability rather than eliminating it.



African Mahogany



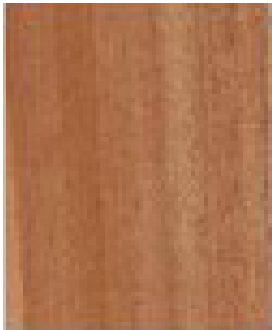
Cherry



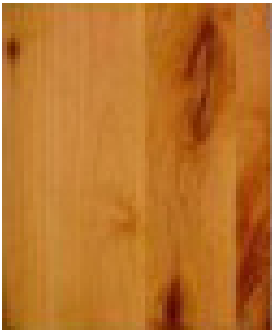
CVG Fir



Hard Maple



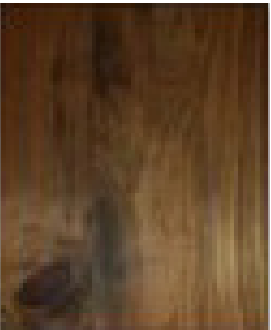
Honduran Mahogany



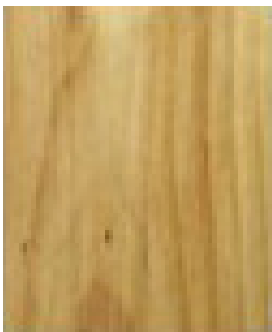
Knotty Alder



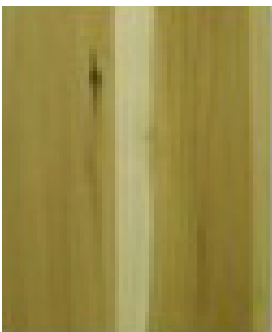
Knotty Pine



Knotty Walnut



Pine



Poplar



Quarter-Sawn
White Oak



Red Oak



Spanish Cedar



Steamed Beech



Walnut



White Ash

Examples of species variation. Of the species shown here, only Fir and Pine are common structural materials

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Wood Structural Basics

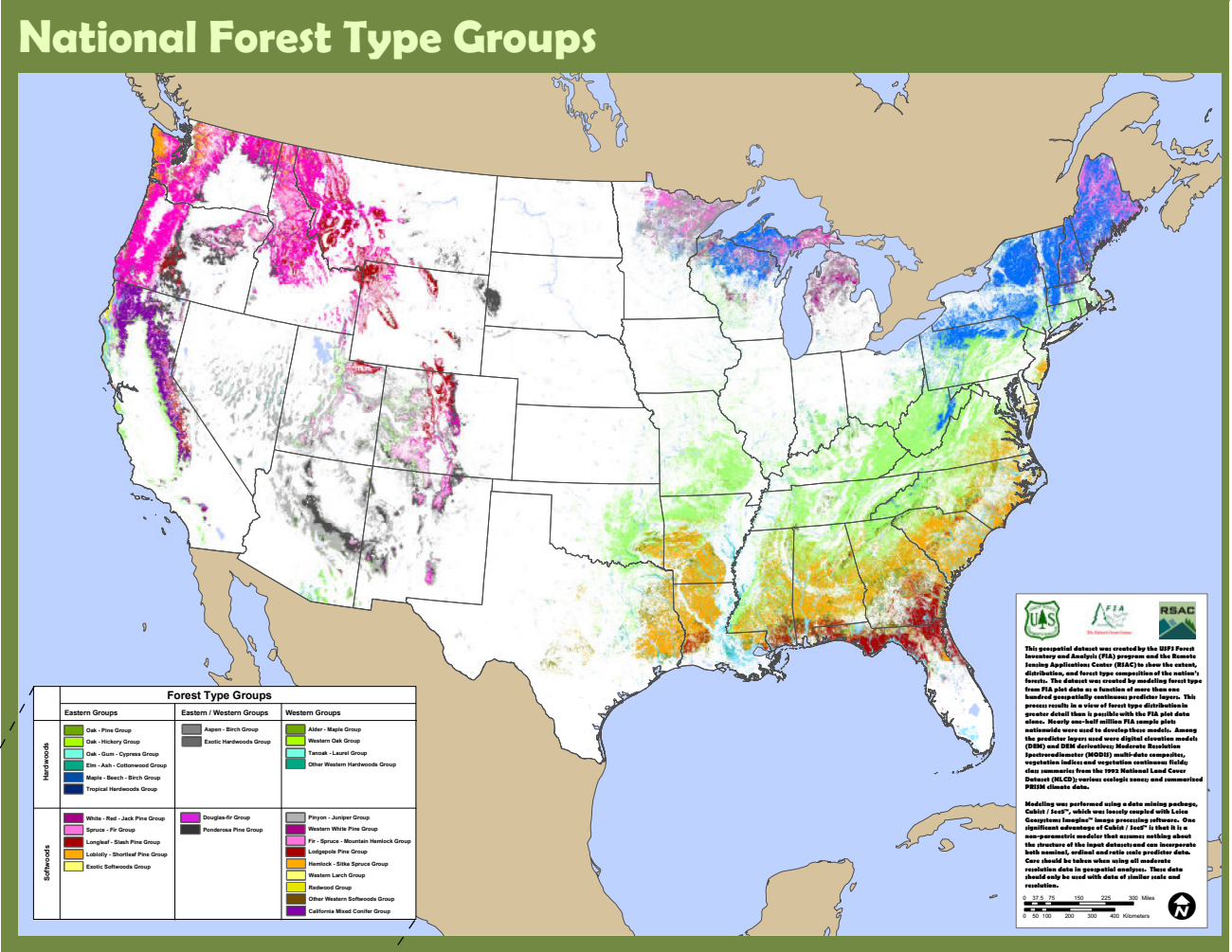
FORESTS

Around the world, and even within the United States, forest composition varies greatly by region. The United States Department of Agriculture (USDA) Forest Service map shows the extent and species distribution of American forests, which are categorized as hardwoods or softwoods, and further differentiated as Eastern, Eastern/Western, and Western groups.

Regional species availability influences material selection in practice, particularly when considering transport distances to mills, manufacturers, fabricators, and the project site. These geographic realities shape which species and products are economical and readily available. Sourcing mass timber from regional supply chains minimizes transportation-related embodied carbon while supporting forest stewardship by creating markets for byproducts of forest management activities such as thinning.

	Forest Type Groups		
	Eastern Groups	Eastern / Western Groups	Western Groups
Hardwoods	Oak - Pine Group Oak - Hickory Group Oak - Gum - Cypress Group Elm - Ash - Cottonwood Group Maple - Beech - Birch Group Tropical Hardwoods Group	Aspen - Birch Group Exotic Hardwoods Group	Alder - Maple Group Western Oak Group Tanoak - Laurel Group Other Western Hardwoods Group
Softwoods	White - Red - Jack Pine Group Spruce - Fir Group Longleaf - Slash Pine Group Loblolly - Shortleaf Pine Group Exotic Softwoods Group	Douglas-fir Group Ponderosa Pine Group	Pinyon - Juniper Group Western White Pine Group Fir - Spruce - Mountain Hemlock Group Lodgepole Pine Group Hemlock - Sitka Spruce Group Western Larch Group Redwood Group Other Western Softwoods Group California Mixed Conifer Group

Enlarged map key



National Forest Type Groups map, USDA Forest Service

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Wood Structural Basics



Douglas Fir Larch (DFL)



Southern Yellow Pine (SYP)



Spruce-Pine-Fir (SPF)
& Spruce-Pine-Fir South (SPF-S)



Hem-Fir (HF)

Common softwood species groupings

SPECIES MATTER, STRUCTURALLY AND AESTHETICALLY

Different wood species have different structural properties. In practice, structural engineers group species with similar structural properties together, like Douglas Fir–Larch (DFL), or Spruce–Pine–Fir (SPF). Some species groups are further differentiated by the region of growth relative to the US-Canadian border (north or south), reflecting differences in strength and stiffness.

Species also vary aesthetically, with noticeable differences in color, texture, grain, and knot frequency. These considerations are critical to architects in unencapsulated mass timber construction, where exposed structural members serve as a finish material.

Mass timber components such as cross-laminated timber (CLT) and glulam are often sourced from different manufacturers, who commonly work with regionally-available species. When visual continuity is a design priority, architects must coordinate species selection across components. For example, CLT in the Pacific Northwest is commonly manufactured from SPF, while glulam is commonly made with DFL.

Wood Structural Basics

SOFTWOODS AND HARDWOODS

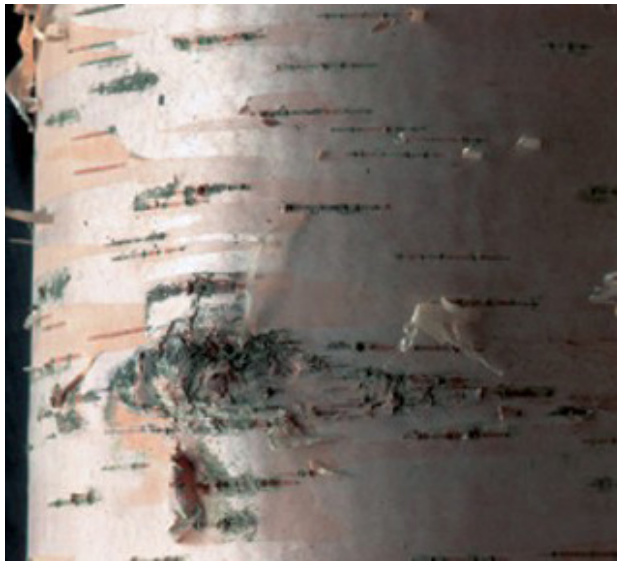
Most structural wood used in buildings is softwood due to its availability, workability, and favorable strength-to-weight ratio. However, there are conditions where the higher density and strength of hardwoods are advantageous, typically in localized, high-stress applications where softwoods cannot meet structural demands with reasonable member sizes.

- Example 1: Beech hardwood is used as dowels in Dowel Laminated Timber (DLT).
- Example 2: BauBuche is a laminated veneer lumber (LVL) product made from beech, a hardwood. It is used in applications requiring high strength, stiffness, or compact member sizing.

The table belows illustrates the relative increase in different properties with hardwood. There is significant variation of these values within each species; values shown are representative.



Douglas Fir, a softwood



Birch, a hardwood

	Specific Gravity	Modulus of Rupture (psi)	Elastic Modulus (psi x 10 ⁶)	Compression Parallel to Grain (psi)	Compression Perpendicular to Grain (psi)	Shear (psi)
Douglas Fir	0.50	12,600	1.83	7,500	800	1,300
Birch	0.65	16,900	2.17	8,500	1,100	2,200
% Increase	130%	134%	119%	113%	138%	170%

Table illustrating relative increase in key material properties when comparing hardwoods to softwoods

Wood Structural Basics



Knot in solid-sawn wood



Glulam

SOLID-SAWN VS ENGINEERED WOOD

Wood is imperfect. While individual wood fibers exhibit very high strength and stiffness, the strength that can be reliably assumed in a structural member is more dependent on the presence and distribution of flaws such as knots, checks, and splits.

Decades of research have gone into the statistical quantification of wood material properties, leading to the classification of wood by grade, which reflects the expected distribution of flaws. Grading may be approximate (visual), based on observable characteristics, or quantifiable (mechanical), using measured structural properties.

The most efficient way to improve the grade of wood is to control the flaw distribution by gluing or mechanically connecting multiple pieces of wood into a single member. This diminishes the impact of individual knots and flaws on a structural member's overall behavior by controlling their distribution. This approach facilitates the use of lower grade wood and smaller diameter trees to produce larger, stronger, more predictable structural members, which in turn reduces reliance on mature, larger diameter trees.

This is the principle benefit of structural composite lumber (SCL) and engineered wood products (EWP), including glulam, plywood, oriented strand board (OSB), laminated strand lumber (LSL), and parallel strand lumber (parallam, PSL).

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Solid-Sawn Lumber Species & Grade	Bending (psi)	Tension (psi)	Shear (psi)	Compression ⊥ to grain (psi)	Compression to grain (psi)	Modulus of Elasticity (psi)	Specific Gravity
Douglas Fir	1,200	800	180	625	1,550	1,800,000	0.50
Southern Yellow Pine	1,650	1,450	175	480	1,800	1,600,000	0.55
Eastern Hemlock	775	350	140	335	1,000	1,000,000	0.36
Southern Yellow Pine	825	500	170	800	825	1,000,000	0.68
Glued Lumber Species & Grade	Bending (psi)	Tension (psi)	Shear (psi)	Compression ⊥ to grain (psi)	Compression to grain (psi)	Modulus of Elasticity (psi)	Specific Gravity
Douglas Fir 24F-1.8E	2,400	1,100	265	650	1,600	1,800,000	0.50
S. Yellow Pine 30F-2.1E	3,000	1,250	300	805	1,750	2,100,000	0.55

Table illustrating relative increase in key material properties when comparing engineered wood to solid-sawn wood

The tables compare key structural properties relevant to structural design for solid-sawn lumber and two common glulam beam grades.

- Example: Douglas Fir glulam has approximately twice the bending strength of solid-sawn Douglas Fir. This significant difference is due to a redistribution of defects rather than a change in base material.

Some properties such as compression strength differ depending on the load direction, parallel or perpendicular to grain. This is called anisotropy.

Wood Structural Basics



Cross-section of wood

ANISOTROPY

Wood is anisotropic: it has unequal properties in different directions. Even a "flawless" piece of wood includes variations in properties depending on the loading direction due to the cellular structure of wood.

A subset of anisotropy is orthotropy, meaning that the differences can be described in terms of two orthotropic (perpendicular) directions. Wood is generally considered orthotropic, with properties specified as parallel or perpendicular to the grain.

To understand orthotropy, think of a tree trunk like a bundle of straws. Straws are quite strong in compression and tension along their length, but crush or pull apart easily if loaded perpendicular the longitudinal axis of the straws. Wood behaves in a similar way.



Bundle of straws

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ANISOTROPY

The diagram shows how allowable stress varies with load direction relative to the grain. Stress is expressed in units of pounds per square inch (psi) in the United States.

- **Compression Parallel to the Grain**
The strongest loading direction. The magnitude of allowable stress in this direction is assigned a relative value of "1"
- **Tension Parallel to the Grain**
Approximately $2/3$ of the allowable parallel compression stress
- **Bending Stress**
Controlled by the same limit state as tension parallel to the grain. Approximately $2/3$ of the allowable parallel compression stress
- **Compression Perpendicular to the Grain**
Approximately $1/3$ of the allowable parallel compression stress
- **Tension Perpendicular to the Grain**
Loading wood in this direction is not allowed. The strength of wood in tension perpendicular to grain is so low that it is assumed to be effectively zero.
- **Shear Stress**
Approximately $1/10$ of the allowable parallel compression stress

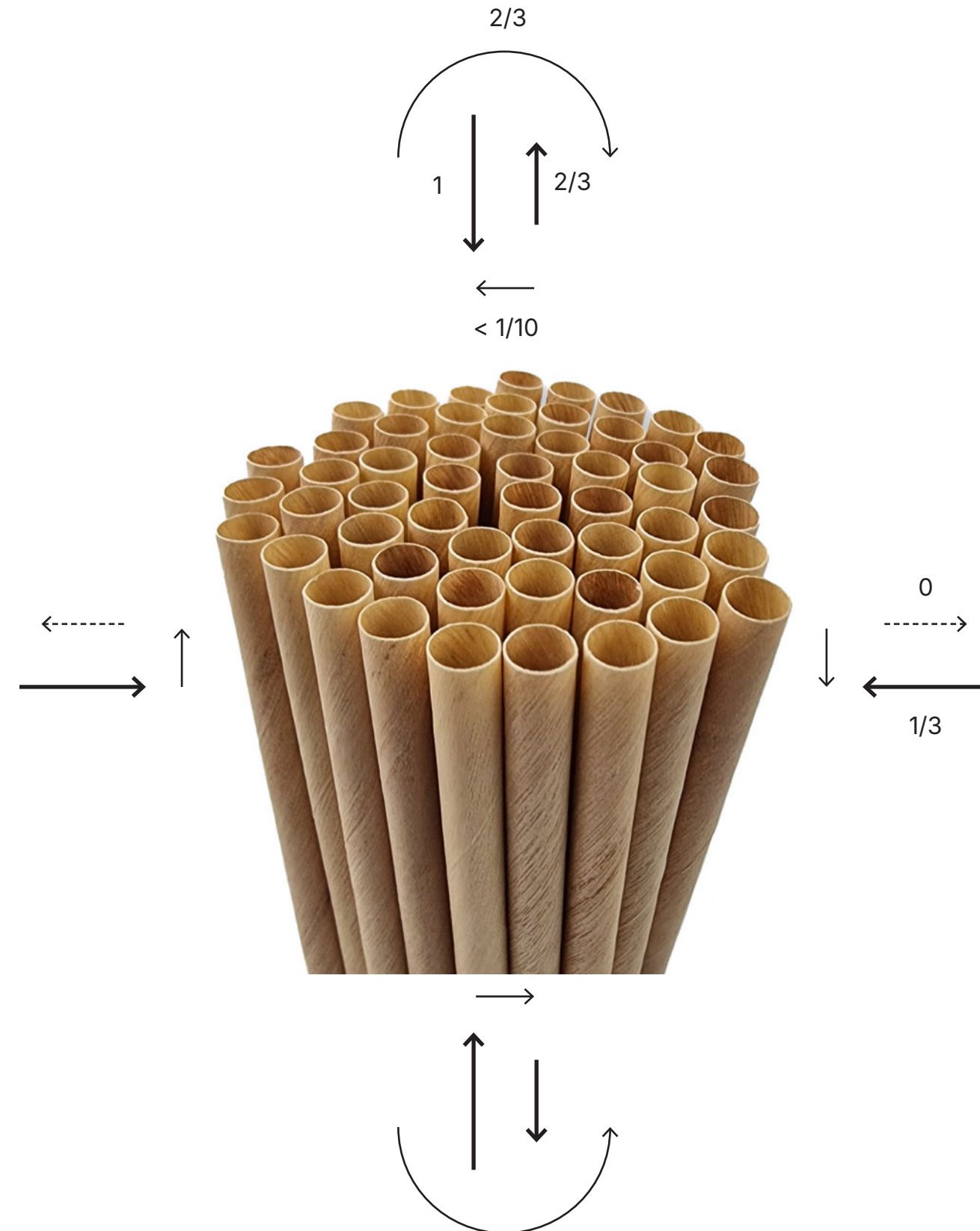
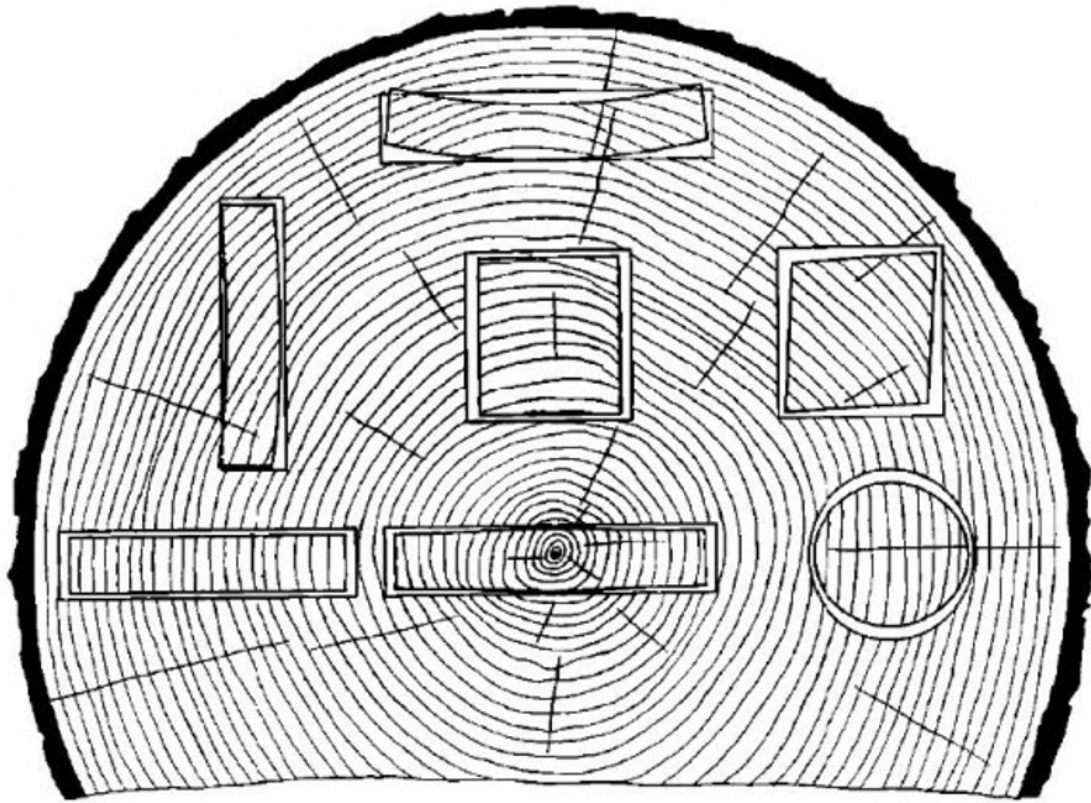


Diagram of relative strength dependent on direction of load relative to the grain

Wood Structural Basics



Sawing pattern and deformations. Lumber cut from different areas of the log will warp, twist, cup, or bow depending on the relationship of the cut to the growth rings

MOISTURE MOVEMENT

One of the most consequential characteristics of wood is its tendency to expand and contract in response to changes in moisture content.

- **Wood shrinks in the process of drying**
 - Freshly cut wood has a high moisture content and will dry over time to its "equilibrium moisture content" (EMC), which depends on local climate and interior conditions, and varies by season
 - Drying may occur naturally over time or be accelerated through kiln drying
- **Wood expands when moisture content increases**
- **Moisture-related movement in wood is directional**
 - Wood shrinks/expands most in the direction tangential to the growth rings
 - Wood shrinks/expands less in the direction radial to the growth rings
 - Wood shrinks/expands negligibly in the direction parallel to the grain (along the axis of the tree trunk)
- **Moisture-related movement is reversible**
 - Wood swells as moisture content increases, and shrinks as it decreases
 - Since grain orientation can vary along the length of a member, movement may be nonuniform and cumulative in larger assemblies

Moisture movement not only influences structural performance, but also both constructibility and long-term durability. Successful timber design requires accommodating movement rather than attempting to prevent it.

Wood Structural Basics



Screw heads protruding through floor panel

MOISTURE MOVEMENT & FASTENERS

Moisture-induced dimensional shrinkage has direct implications for connections and detailing. Even in engineered wood products, the wood changes in volume more perpendicular than parallel to grain.

As wood dries and shrinks, fasteners installed at higher moisture content may become exposed as the surrounding material recedes.

- Example 1 (top image): A line of screw heads installed when the wood was at a higher moisture content is now protruding above the floor panel, through the flooring
- Example 2 (bottom image): A gap between screw heads and a steel plate due to shrinkage. In contrast, when wood swells, a fastener may be placed in tension, stretching due to wood swelling, possibly resulting in fracture



Gap between screw heads and steel plate

Mass Timber Structural Components

MASS TIMBER FRAMING OPTIONS

Mass timber construction employs a variety of structural elements, including panels, beams, and columns assembled into floor, wall, and roof systems.

Though panelized products, particularly CLT, have surged in popularity since the 2010s, mass timber encompasses a broader family of panel and lineal elements with distinct structural capabilities and aesthetic outcomes. These systems differ in appearance, maximum span, structural depth, dimensional stability, and response to moisture movement, making material selection a critical early design decision.



Cross-Laminated Timber (CLT)



Nail-Laminated Timber (NLT)



Glulam Beams & Columns



Glue-Laminated Timber (GLT)

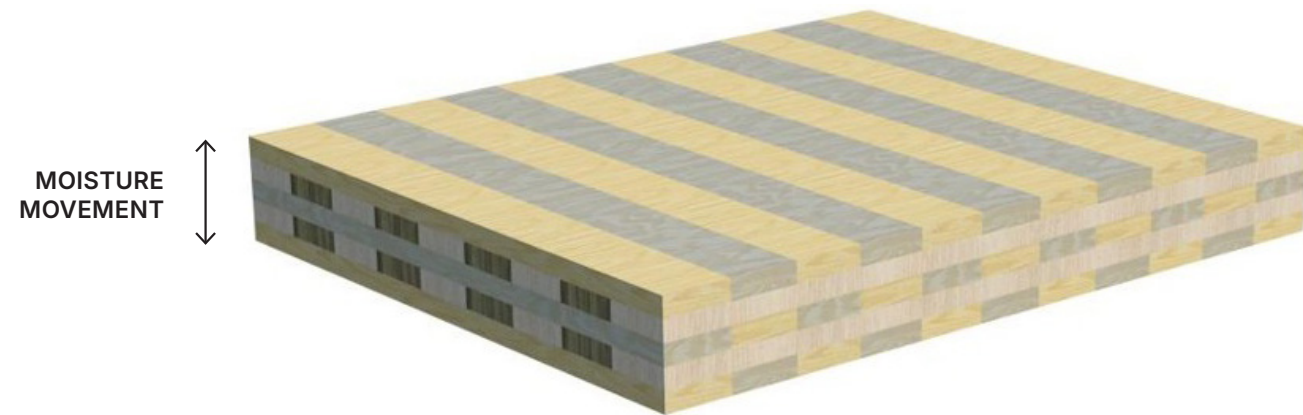


Laminated Veneer Lumber (LVL)

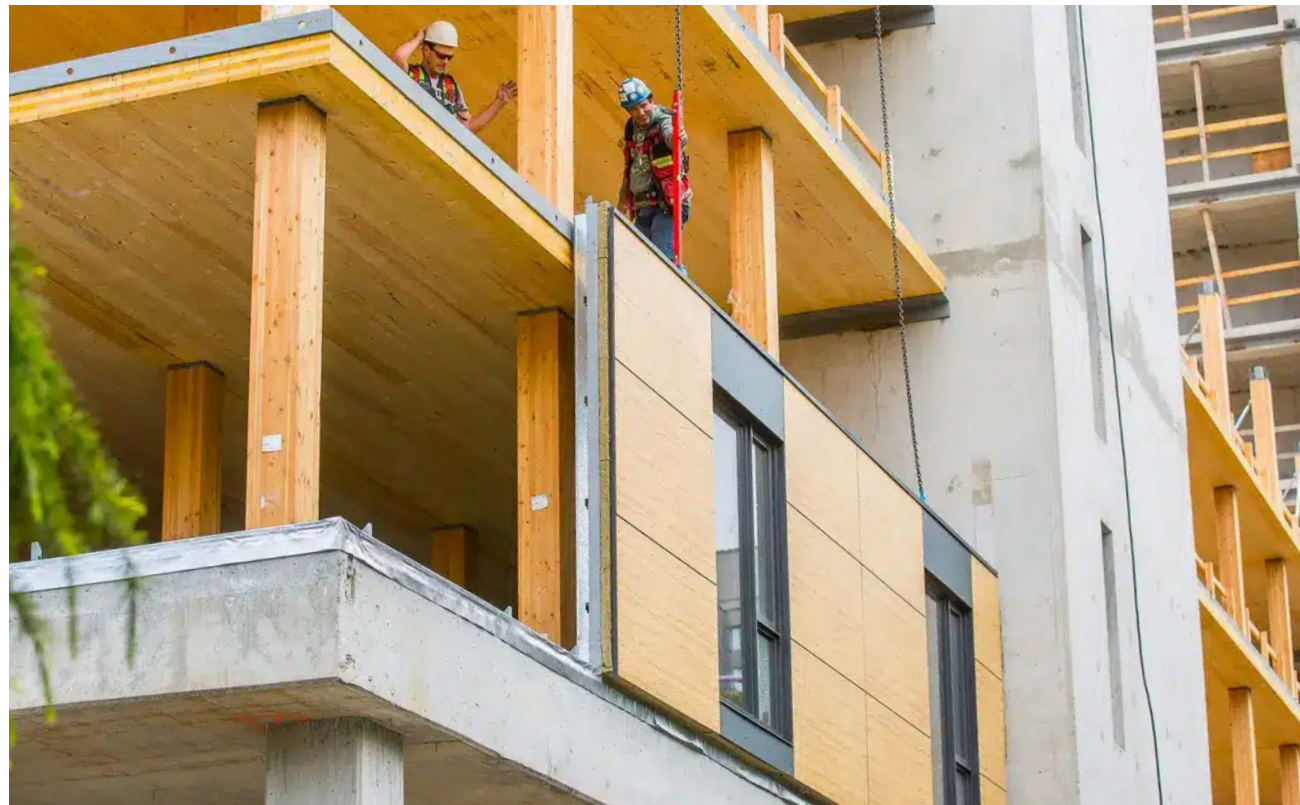
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CLT Layup



Brock Commons Tallwood House, Vancouver, BC, Canada, 2017; Architect: Acton Ostry Architects, Structural Engineer: Fast + Epp

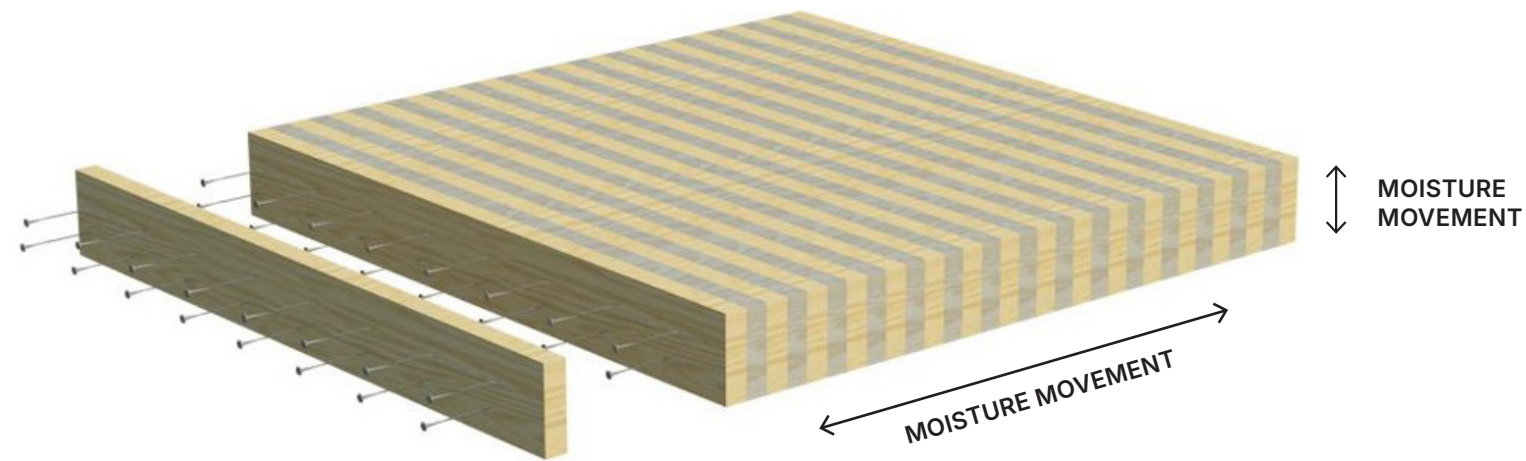
CROSS-LAMINATED TIMBER (CLT)

- Most common mass timber panel
- Manufactured from dimensional lumber (typically 2x) glued into multiple layers
- Adjacent layers are oriented in alternating (perpendicular) directions
- Some two-way spanning capability
- Can be used for floors, walls, or roofs
- Standardized product manufactured in North America, South America, Europe, Asia, and Oceania
- Moisture movement is limited to minor shrinkage or expansion in depth because alternating grain directions of adjacent plies provide dimensional stability in plane, similar to plywood. This is one of the significant advantages of CLT (and MPP) over other mass timber panels

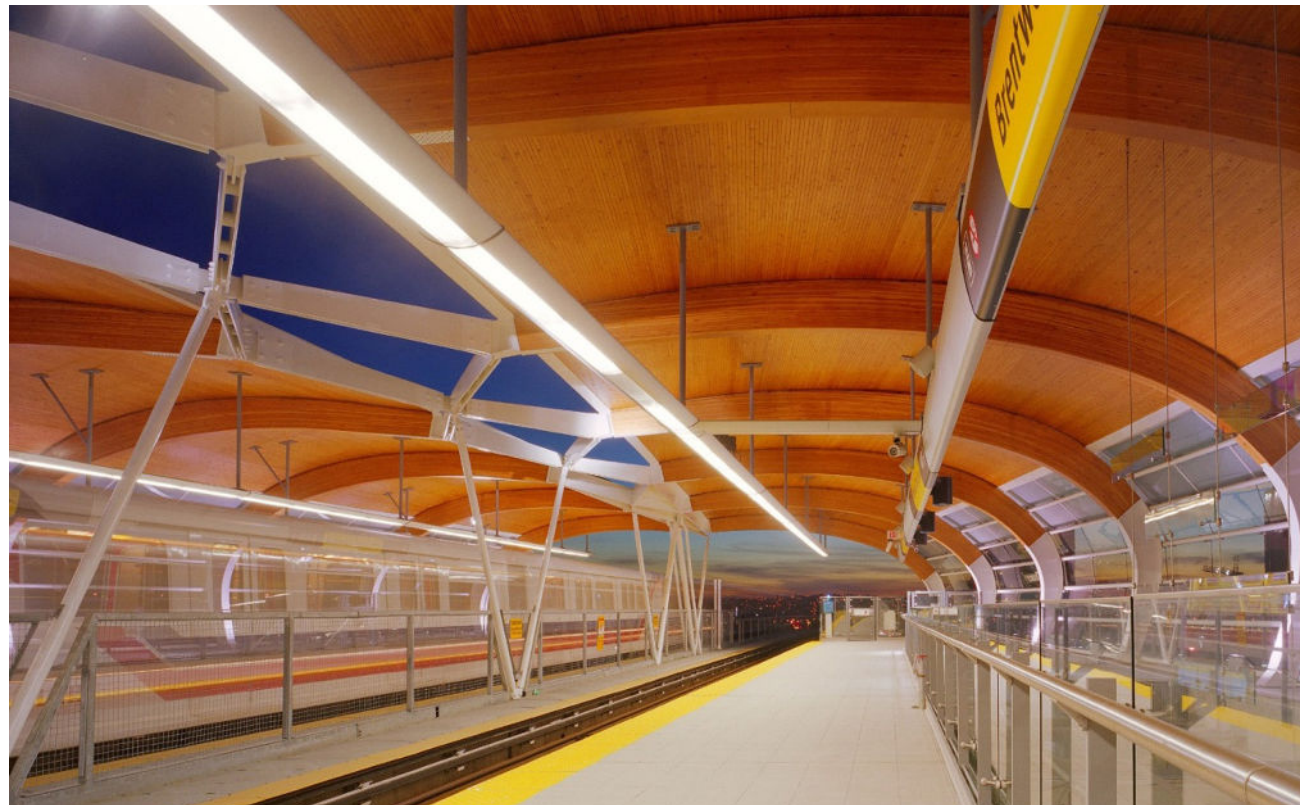
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NLT Member Build-Up



Brentwood Station, Burnaby, BC, Canada, 2002; Architect: Perkins&Will, Structural Engineer: Fast + Epp

NAIL-LAMINATED TIMBER (NLT)

- Similar to DLT
- Manufactured from dimensional lumber (typically 2x or 3x) installed on edge in a single layer
- Lumber fastened together with nails made of steel, aluminum, or wood
- One-way spanning system
- Requires structural sheathing such as plywood to function as a diaphragm
- Can be used for floors, walls, and roofs
- Non-standardized panel system
- Base lumber governed by typical grading rules, so structural properties of panels without butt joints can be determined through basic engineering principles
- Design guides available for the U.S. and Canada
- Can be supplied by any carpenter
- Panels can be prefabricated or built in place
- Low tech and low cost compared to glued and laminated elements
- Moisture movement can be substantial due to cumulative movement of individual members. Significant movement joints are required

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DOWEL-LAMINATED TIMBER (DLT)

- Similar to NLT
- Manufactured from dimensional lumber (typically 2x or 3x) installed on edge in a single layer
- Lumber fastened together with hardwood dowels. This allows DLT to be cut and modified more easily than NLT
- One-way spanning system
- Requires structural sheathing such as plywood to function as a diaphragm
- Can be used for floors, walls, and roofs
- Non-standardized panel system
- Base lumber governed by typical grading rules, so structural properties of panels without butt joints can be determined through basic engineering principles
- Panels must be prefabricated. Specialized fabrication process makes it more complex than NLT
- Low cost compared to glued and laminated elements
- Moisture movement can be substantial due to cumulative movement of individual members. Significant movement joints are required

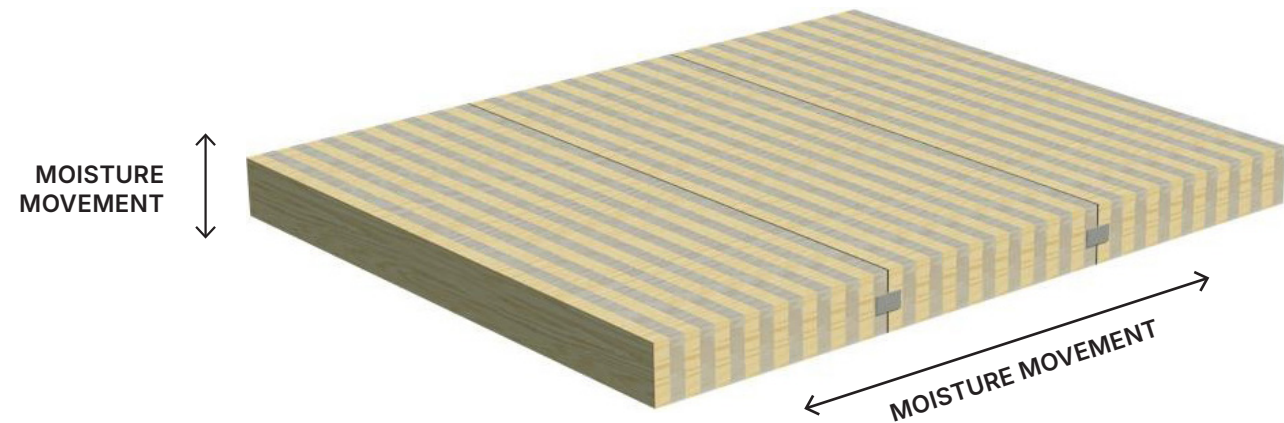


The Soto, San Antonio, TX, 2020; Architect: Lake|Flato Architects, BOKA Powell, Structural Engineer: StructureCraft

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GLT Layup



Kin Centre Complex, Prince George, BC, Canada, 2013; Architect: PBK Architects, Inc, Structural Engineer: Fast + Epp

GLUE-LAMINATED TIMBER (GLT)

- Similar to a Glulam Column or Beam on its side, but uniformly graded
- Similar to NLT and DLT with regards to spanning direction, moisture movement, and structural applications
- Manufactured from dimensional lumber (typically 2x) installed on edge and glued together in a single layer
- One-way spanning system
- Requires structural spline to function as a diaphragm
- Can be used for floors, walls, and roofs
- Typically sheathed with plywood for floor applications
- Standardized product, manufactured in North America, South America, Europe, Asia, and Oceania
- Moisture movement can be substantial due to cumulative movement of individual members. Significant movement joints are required

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Glulam Layup



Haesley Hamlet Grand Hall, Yeosu, Korea, 2020; Architect: Shigeru Ban Architects, Structural Engineer: Création Holz

GLULAM BEAMS AND COLUMNS

- Similar to GLT, but non-uniformly graded
- Manufactured from dimensional lumber (typically 2x). Commonly glued together in either a single or multi-lamination width (2 or more laminations wide)
- Laminations arranged strategically to place higher-grade material in zones of highest stress
- One-way spanning system
- Can be used for lineal elements such as beams and columns
- Non-standardized product. Grade for each lamination is controlled by specific demand and structural configuration
- Moisture movement is the same as for solid-sawn timber of similar size. Checking occurs, but is inhibited by discontinuity of grain at laminations

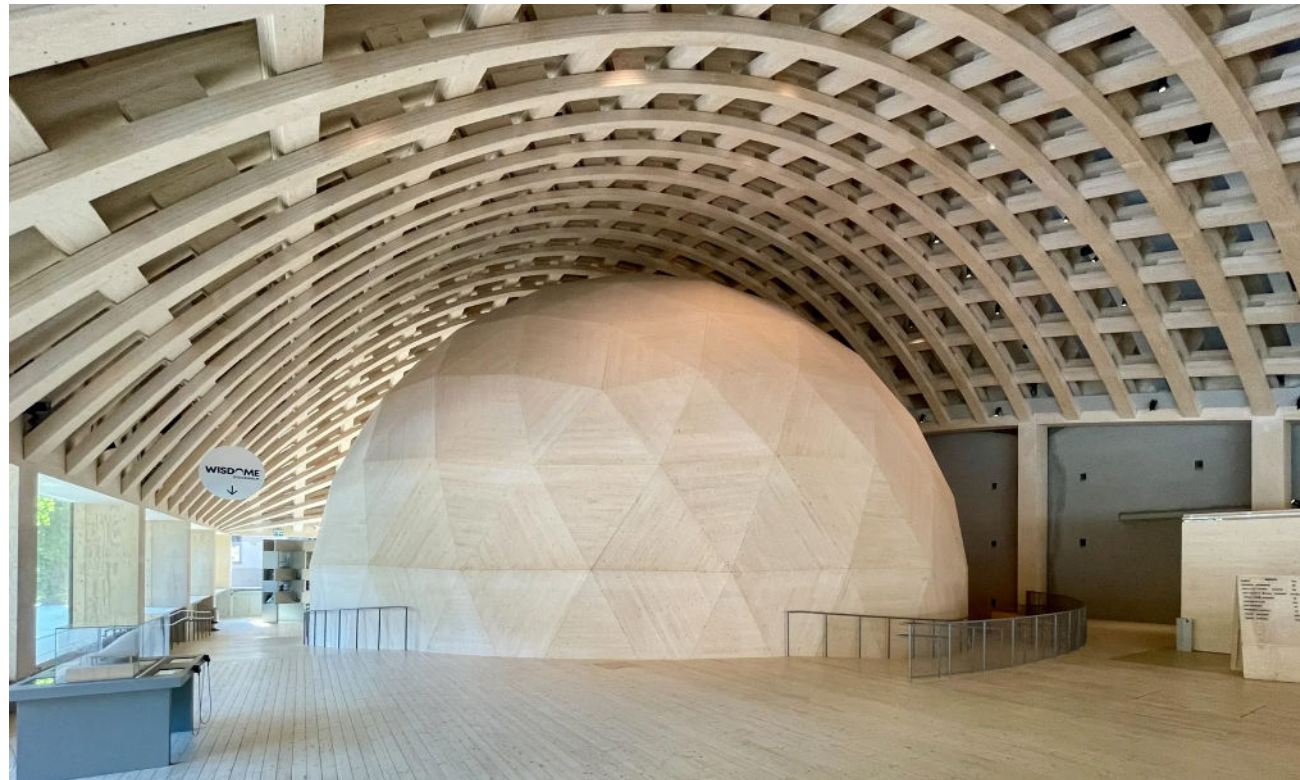
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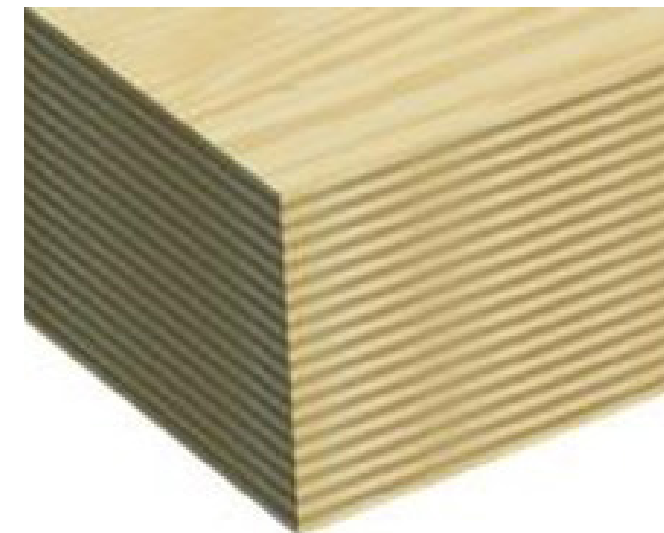
LVL Layup



LVL roof structure: Wisdom Stockholm, Stockholm, Sweden, 2023;
Architect: Elding Oscarson, Structural Engineer: Création Holz, SJB Kempter Fitze, Florian Kosche (Dipl.-Ing.)

LAMINATED VENEER LUMBER (LVL)

- Similar to plywood, but grain orientation of all laminations in the same direction
- Higher strength and stiffness than plywood
- Manufactured from wood veneers stacked horizontally and glued together
- One-way spanning system
- Can be used for floors, walls, and roofs
- Standardized product, manufactured in North America, South America, Europe, Asia, and Oceania
- Moisture movement most significant in thickness
- Relatively dimensionally stable in plane

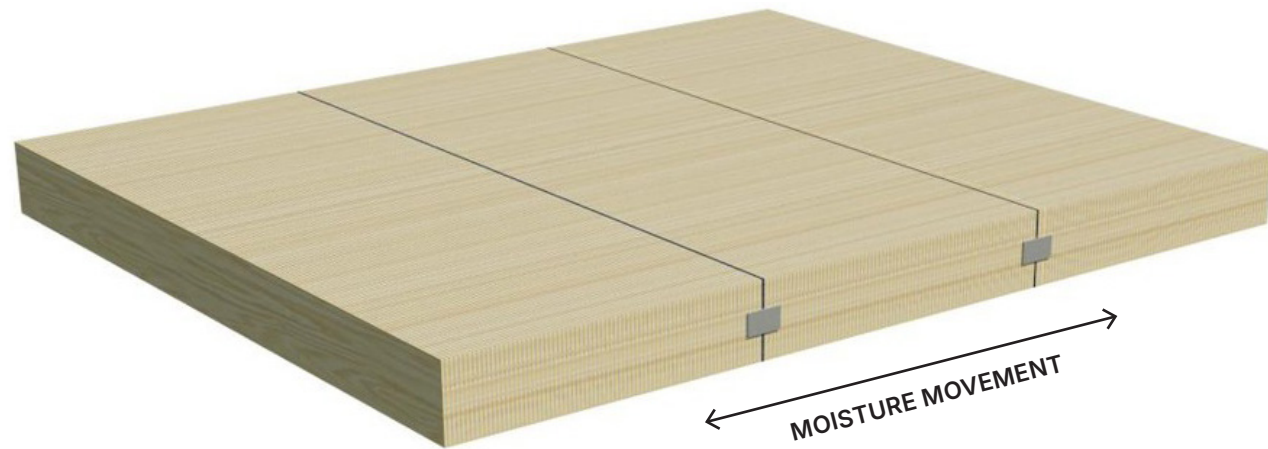


Detail of LVL assembly

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Secondary LVL Layup

SECONDARY LAMINATED VENEER LUMBER

- Manufactured from wood veneers stacked vertically and glued together
- One-way spanning system
- Can be used for floors, walls, and roofs
- Standardized product, manufactured in North America, Europe, and Oceania
- Sometimes referred to by supplier names (such as Brisco)
- Moisture movement most significant in-plane, perpendicular to the grain
- Dimensionally stable in thickness



Shoreline, Vancouver, BC, Canada, 2014; Architect: Michael Green Architecture, Structural Engineer: Equilibrium Consulting Inc.



Detail of secondary LVL assembly

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LSL Panel Composition

LAMINATED STRAND LUMBER (LSL)

- Manufactured from wood strands and chips bonded with adhesive
- Grain is oriented somewhat randomly, but in a primary direction
- One-way spanning system
- Can be used for floors, walls, and roofs
- Standardized product, manufactured in North America, with limited production in Europe
- Moisture movement most significant in thickness
- Relatively dimensionally stable in plane



Gilmore SkyTrain Station, Burnaby, BC, Canada, 2002; Architect: Busby and Associates, Structural Engineer: Fast + Epp

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MPP Layup

MASS PLYWOOD PANELS (MPP)

- Similar to plywood, scaled up to building-scale structural components
- Manufactured from wood veneer glued into multiple layers
- Adjacent layers are oriented in alternating (perpendicular) directions
- Two-way spanning system
- Can be used for floors, walls, and roofs
- Standardized proprietary product, only manufactured in North America
- Patented by Freres Engineered Wood
- Manufactured in thicknesses up to 18" and lengths up to 80'
- Strong, stiff, and versatile
- Dimensionally stable in plane. Moisture movement in thickness only, similar to CLT



Freres MPP Warehouse, Mill City, Oregon, 2024; Architect & Structural Engineer: Crow Engineering

Grading

CLT performance is dictated by the species and grade of the substrate wood, which can vary between longitudinal and transverse layers within the same panel. Because wood is orthotropic, properties such as bending, shear, and compression are dependent on the layer orientation.

With variation among plies, grading affects both structural capacity and visual appearance, which is of particular importance in exposed applications.

VISUAL CLT GRADES

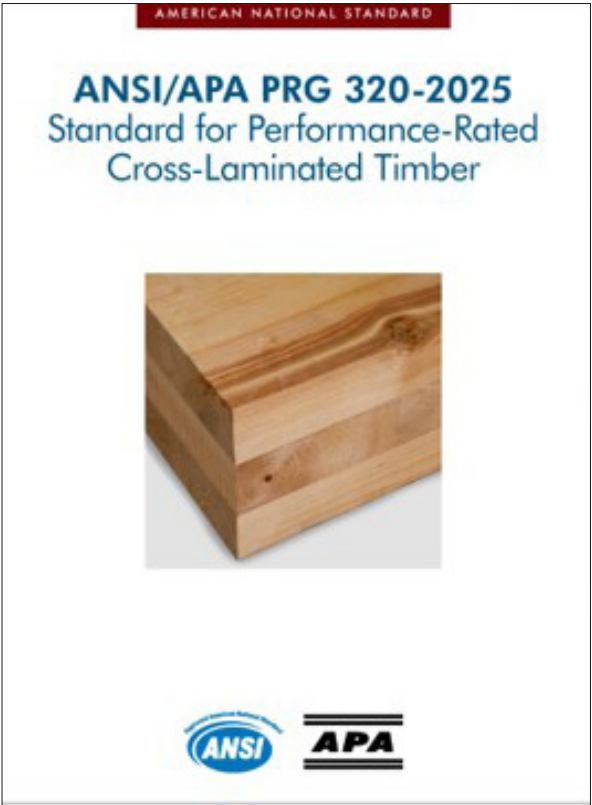
Visual CLT grades define standardized CLT panel layups using visually graded lumber, where structural properties reflect statistically quantified expectations of knots and other visible defects.

Designations V1-V5 are generic layups defined in the manufacturing standard ANSI/APA PRG 320-2025.

The designation (for example, V4) does not represent a quality ranking or finish grade: the number indicates species and grade combinations in longitudinal and transverse layers.

Designation	Direction	Grade	Species
V1	Longitudinal Transverse	No 2 No 3	Doug Fir Larch Doug Fir Larch
V1 (N)	Longitudinal Transverse	No 2 No 3	Doug Fir Larch North Doug Fir Larch North
V2	Longitudinal Transverse	No 1 / No 2 No 3	Spruce Pine Fir Spruce Pine Fir
V3	Longitudinal Transverse	No 2 No 3	Southern Pine Southern Pine
V4	Longitudinal Transverse	No 2 No 3	Spruce Pine Fir South Spruce Pine Fir South
V5	Longitudinal Transverse	No 2 No 3	Hem Fir Hem Fir

Annex A — Design Properties for ANSI/APA PRG-320 CLT



Manufacturing standard for CLT in the United States

Grading

MECHANICALLY STRESS-RATED LUMBER GRADES IN CLT

Mechanically stress-rated (MSR) lumber grades define standardized CLT panel layups using non-destructive lumber testing rather than visual inspection.

Designations E1-E5 are generic layups defined in ANSI/APA PRG 320-2025. As with visual grades, the designation (for example, E4) describes a specific species and grade combination in longitudinal and transverse layers.

Designation	Direction	Grade	Species
E1	Longitudinal Transverse	1950f 1.7E No 3	Spruce Pine Fir MSR Spruce Pine Fir
E2	Longitudinal Transverse	1650f 1.5E No 3	Doug Fir Larch MSR Doug Fir Larch
E3	Longitudinal Transverse	1200f 1.2E No 3	Eastern, Northern, Western Softwoods MSR Eastern, Northern, Western Softwoods
E4	Longitudinal Transverse	1950f 1.7 E No 3	Southern Pine MSR Southern Pine
E5	Longitudinal Transverse	1650f 1.5E No 3	Hem Fir MSR Hem Fir

Annex A — Design Properties, ANSI/APA PRG-320 CLT

MECHANICALLY STRESS-RATED VENEER GRADES IN EWPs

MSR veneer grades define layups that incorporate engineered veneer-based structural composite lumber (SCL) such as laminated veneer lumber (LVL), laminated strand lumber (LSL), and oriented strand lumber (OSL) within CLT.

Designations S1-S3 are generic layups defined in ANSI/APA PRG 320-2025 which define orientation, veneer grade, and structural component type. The use of engineered wood lamellas enables higher and more uniform material properties in selected layers, including increased strength and stiffness.

Designation	Direction	Grade	Structural Composite Lumber (SCL) Type
S1	Longitudinal Transverse	2250f 1.5E 2250f 1.5E	LVL LVL
S2	Longitudinal Transverse	1900f 1.3E 1900f 1.3E	LSL LSL
S3	Longitudinal Transverse	1750f 1.3E 1750f 1.3E	OSL OSL

Annex A — Design Properties, ANSI/APA PRG-320 CLT

Mass Timber Structural Systems

Mass timber structural systems organize panels, beams, and columns into load-bearing configurations.

The choice of structural system depends on a variety of factors, such as building program, span requirements, layout flexibility, facade apertures, and overall architectural intent.



Diagram of T3 Minneapolis structure, featuring NLT panels and glulam columns and beams



T3, Minneapolis, MN, 2016; Architect: Michael Green Architecture, Structural Engineer: Magnusson Klemencic Associates

**STRUCTURAL
MODULE 1**

**Structural
Basics of
Mass Timber**

Mass Timber Structural Systems

BEARING WALL SYSTEMS

- Vertical loads carried primarily by walls
- Walls may be mass timber (CLT, NLT, DLT, MPP) or light wood framing
- Efficient load path
- Commonly used for regular plans and stacked layouts
- Suitable for occupancies requiring spatial separation, such as schools and residential buildings



Mississippi Workshop, Portland, OR, 2022; Architect: Waechter Architecture, Structural Engineer: KPFF Consulting Engineers



The Duke, Austin, TX, 2021; Architect: DesignTrait, Structural Engineer: Ava Lamb



6-CLT-Story Building, Modena, Italy, 2017; Architect: Fabbriart, Structural Engineer: Ergodomus Timber Engineering

STRUCTURAL MODULE 1

Structural Basics of Mass Timber

Mass Timber Structural Systems

POST-AND-PLATE (POINT-SUPPORTED) SYSTEMS

- Vertical loads carried by columns supporting flat panels
- Typically uses CLT or other two-way spanning panel
- Efficient structural depth, with no beams, girders, or joists interrupting ceiling
- Compatible with large windows
- Limitations
 - Column spacing must align with panel dimensions
 - Two-way spanning panels require greater thickness than one-way systems

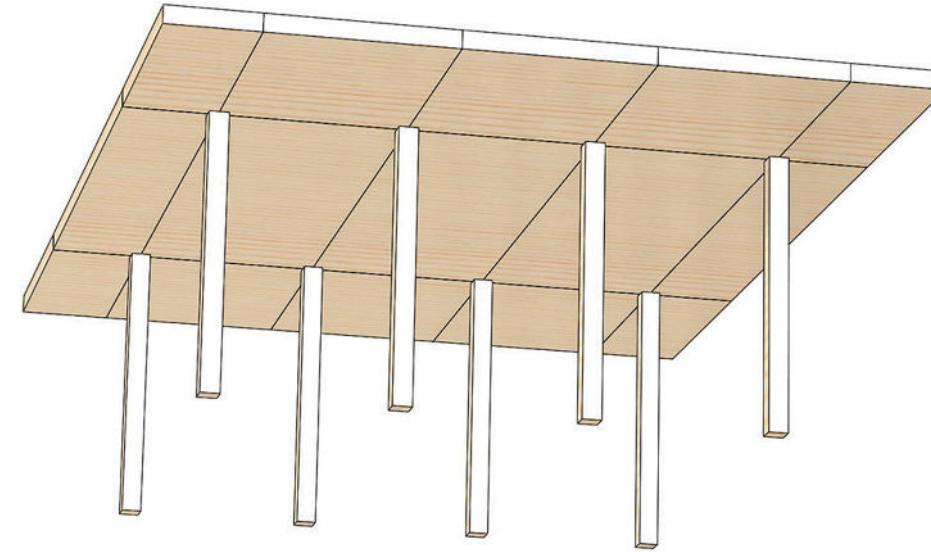
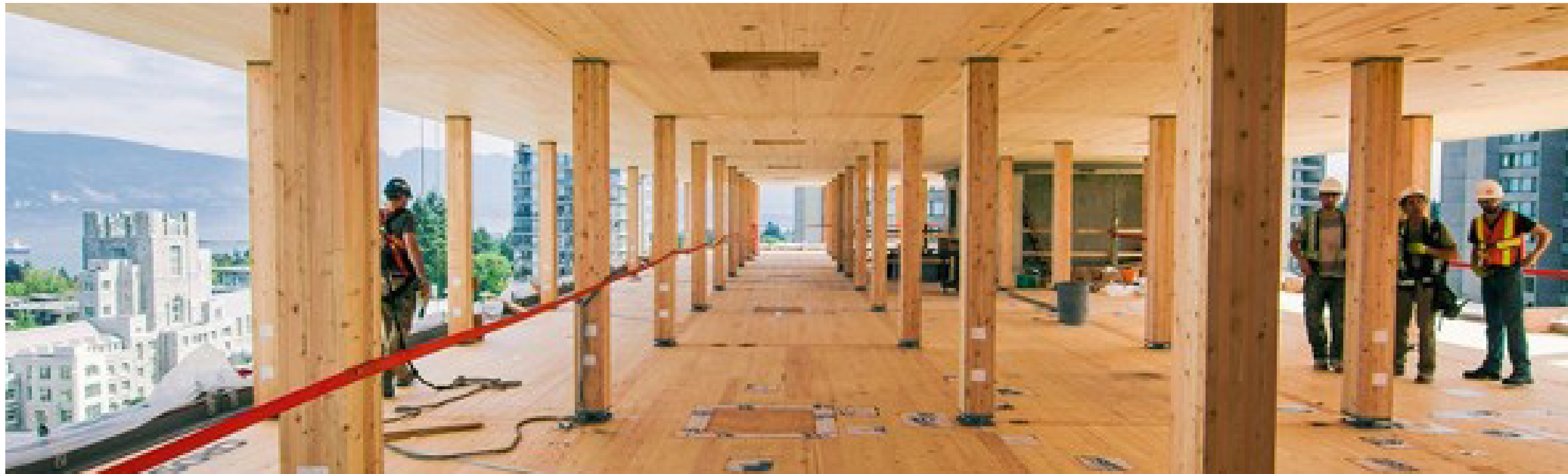


Diagram of post-and-plate system



Brock Commons Tallwood House, Vancouver, BC, Canada, 2017; Architect: Acton Ostry Architects, Structural Engineer: Fast + Epp

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Basics of
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Mass Timber Structural Systems

POST-AND-BEAM (FRAME) SYSTEMS

- Vertical loads carried by beams spanning between columns
- Floors typically constructed from one-way spanning panels or secondary framing
- Flexible structural system with many possible configurations
- Commonly used for open plans and long span
- Compatible with large windows
- Suitable for commercial, office, and assembly occupancies

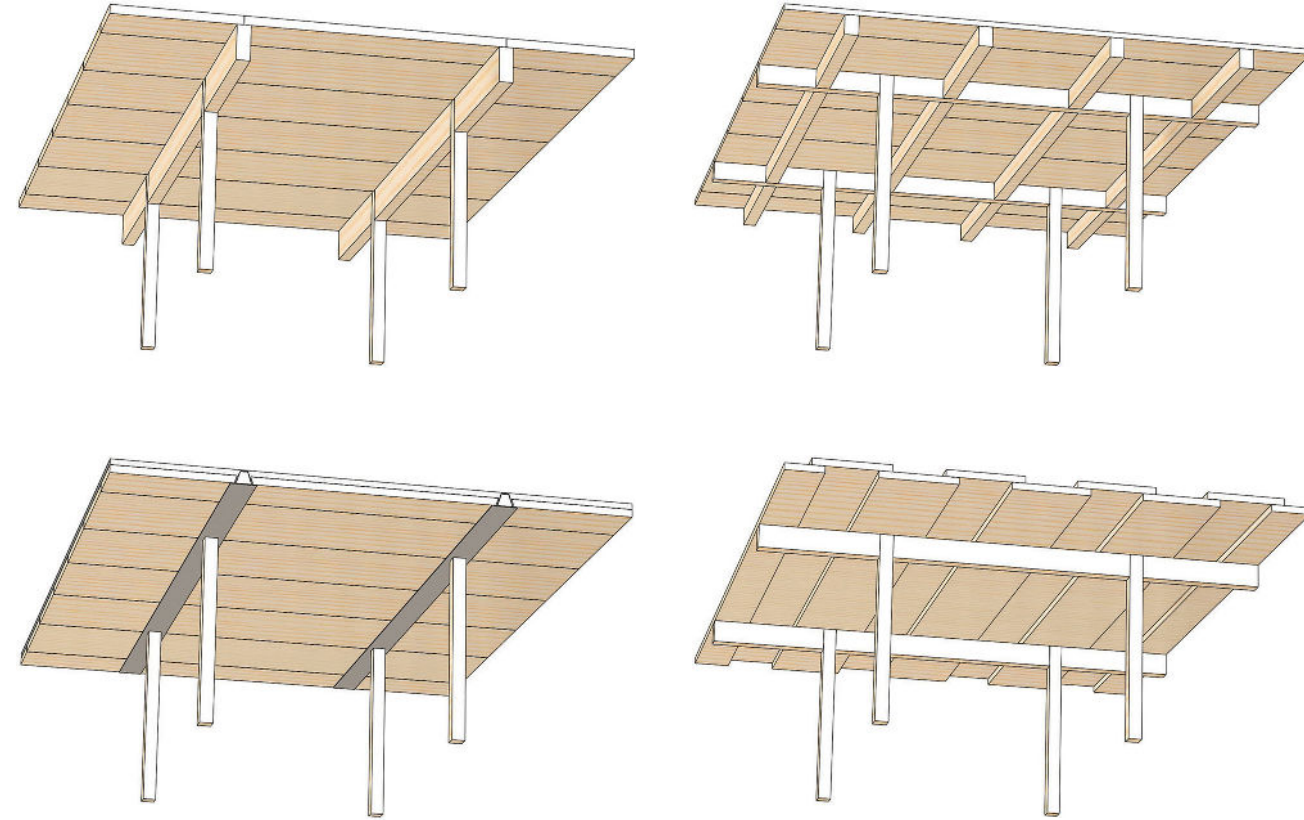


Diagram of selected post-and-beam systems



Northlake Commons, Seattle, WA, 2024; Architect: Weber Thompson, Structural Engineer: DCI Engineers

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Mass Timber**



Hybrid Structural Systems

Hybrid mass timber construction combines mass timber with another structural material such as steel, concrete, or light wood framing.

Hybrid approaches are often employed to address specific requirements, such as long spans, lateral resistance, fire performance, acoustical separation, structural efficiency, or cost. While fully mass timber structures are possible, they are not always the most efficient approach to a given project.

Building codes and regulatory requirements frequently drive hybrid solutions. For example, the use of unencapsulated mass timber may be restricted in certain occupancies, building heights, or floor areas, making hybrid systems a practical way to achieve compliance without sacrificing the benefits of mass timber.

Hybrid structural systems are not a compromise: they provide opportunities to expand the architectural and structural possibilities of mass timber buildings.



Diagram of hybrid mass timber system at the Adidas Headquarters Expansion and Gold Building, featuring precast concrete columns and girders, glulam beams, and CLT panels

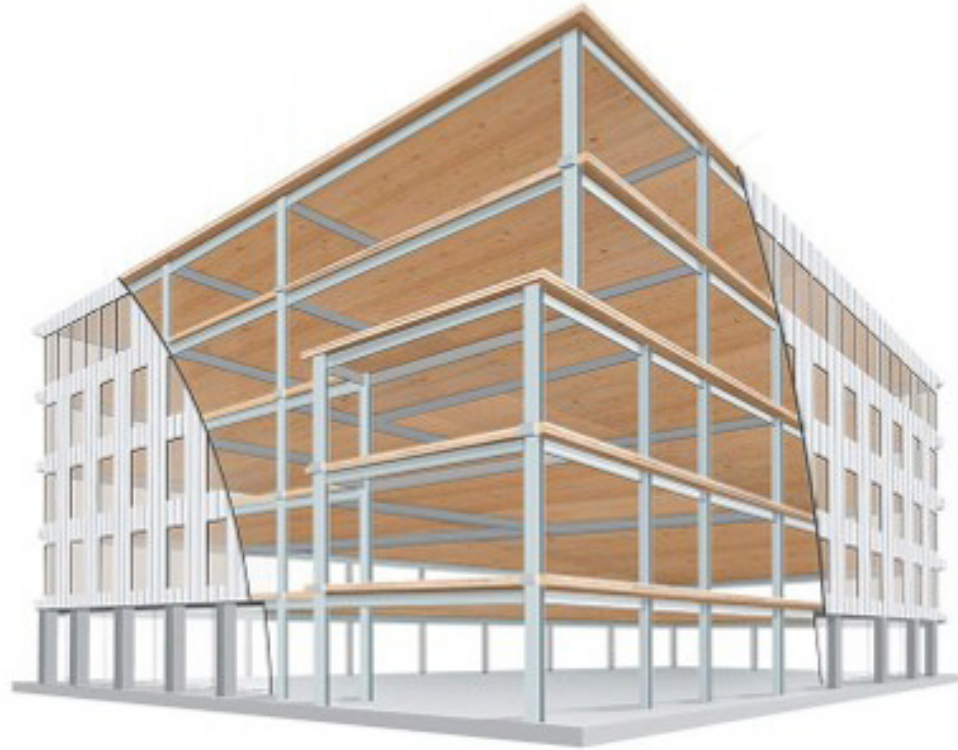


Adidas Headquarters Expansion and Gold Building, Portland, OR, 2021; Architect: LEVER Architecture, Structural Engineer: KPFF Consulting Engineers

STRUCTURAL MODULE 1

Structural Basics of Mass Timber

Hybrid Structural Systems



Cutaway diagram of mass timber and steel structure

MASS TIMBER AND STEEL

The incorporation of steel facilitates:

- Long spans
- Shallow structural depth
- Coordination of mechanical, electrical, and plumbing systems
- Reuse of steel components to promote circular economy strategies
- Fire-rated building envelope systems



Fire Station No.3, Boulder, CO, 2024; Architect: Davis Partnership Architects, Structural Engineer: KL&A Engineers and Builders



Fire Station No.3, Boulder, CO, 2024; Architect: Davis Partnership Architects, Structural Engineer: KL&A Engineers and Builders

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Basics of
Mass Timber**

Hybrid Structural Systems



Cutaway diagram of mass timber and concrete structure

MASS TIMBER AND CONCRETE

The incorporation of concrete facilitates:

- Lateral load resistance through cores and shear walls
- Structural transfers between podium and tower levels
 - Example: Office or residential floors above retail or amenity spaces
- Achieving fire resistance ratings and acoustic separation between occupancies (vertically)
- Parking levels above and below grade



Platte Fifteen, Denver, CO, 2020; Architect: OZ Architecture, Structural Engineer: KL&A Engineers and Builders

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Basics of
Mass Timber

Hybrid Structural Systems



Cirrus, Denver, CO, United States, 2022; Architect: Keith Boyd, Structural Engineer: KL&A Engineers and Builders

MASS TIMBER AND WOOD LIGHT FRAME

The incorporation of light wood framing facilitates:

- Prefabricated and panelized construction
- Lower-cost bearing wall systems than mass timber
- Wall systems with integrated plumbing and electrical systems
- Non-structural infill walls in residential and mixed-use buildings



Virtuoso, Vancouver, BC, Canada, 2018; Architect: Rostich Hemphill Architects, Structural Engineer: Wicke Herfst Maver

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Fire Resistance

FIRE PERFORMANCE

Fire performance is a key consideration for any timber structure. A common misconception is that wood structures are not safe because they are combustible, when in actuality, they must comply with code requirements that are comparable to or exceed the fire safety requirements for other materials.

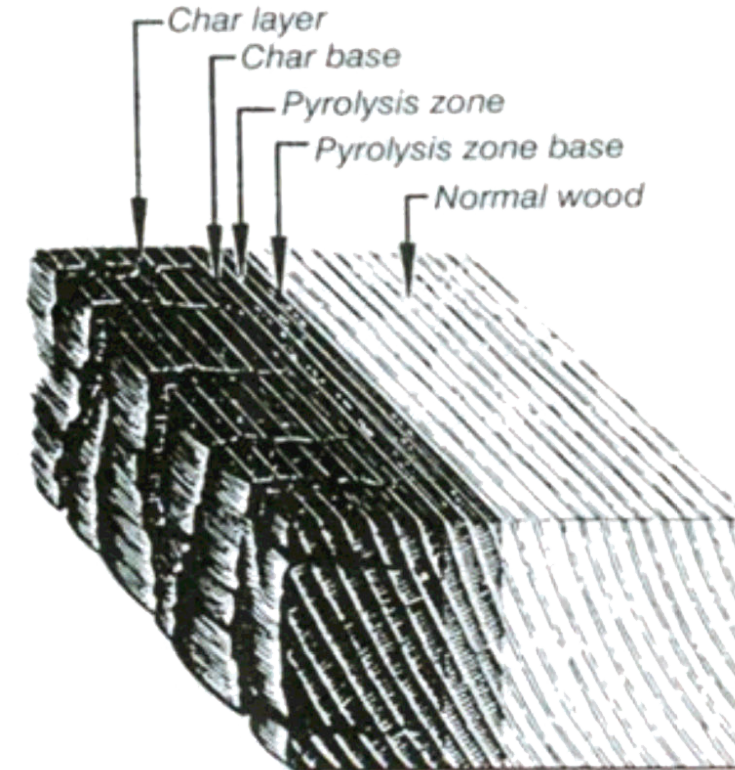
Dimensional lumber burns quickly due to its small cross section and therefore almost always requires encapsulation with a non-combustible material such as gypsum board to protect it.

In contrast, mass timber and heavy timber members have inherent fire resistance due to their thickness. When exposed to fire, the outer layer burns and forms a thick, protective char that insulates the inner wood fibers and slows the burn to a predictable rate of approximately 1.5" per hour.

The building code for Type IV construction recognizes this characteristic and establishes guidelines for calculating fire-resistance ratings (FRR) and member sizing to carry live and dead loads even after charring.



3-hour rated fire testing of a glulam column



Degradation zones in wood exposed to fire

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Fire Resistance

FIRE-RESISTANCE RATINGS

Fire performance is governed by fire-resistance rating (FRR), which is defined in the 2024 International Building Code (IBC) as “the period of time a building element, component or assembly maintains the ability to confine a fire, continues to perform a given structural function, or both, as determined by the tests, or the methods based on tests, prescribed in Section 703.”

Fire resistance requirements vary by construction type and structural element, and can be primarily categorized into rated and unrated construction. FRR is expressed as a duration, such as a 1-hour fire rating.

In some cases, the required FRR cannot be achieved through timber member sizing alone. The fire resistance of timber can be combined with noncombustible encapsulation, such as gypsum board, to achieve the required FRR. While this approach can be used to meet code requirements, it may conflict with design goals that prioritize exposed mass timber.



Unrated light-frame timber structure



2-hour fire rated mass timber structure

ENCAPSULATION

Required Fire Rating	Minimum Protection Required from Noncombustible Protection
1 Hour	40 min 1 layer 5/8" Type X
2 Hour	80 min 2 layers 5/8" Type X
3+ Hour	120 min 3 layers 5/8" Type X



Encapsulated mass timber columns:
Brock Commons Tallwood House, Vancouver, BC, Canada, 2017;
Architect: Acton Ostry Architects, Structural Engineer: Fast + Epp

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Heavy timber structure

MINIMUM SIZE REQUIREMENTS FOR TYPE IV CONSTRUCTION

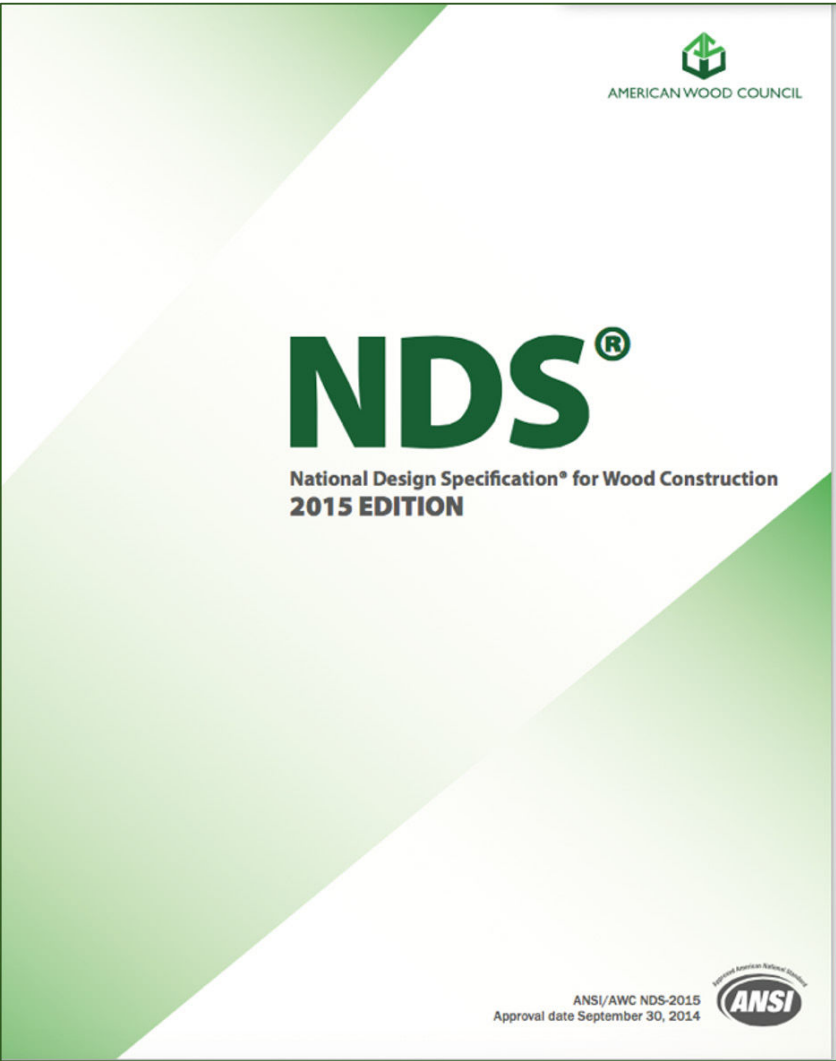
Mass timber elements in Type IV construction must meet the same minimum member size requirements that historically defined Type IV-HT (heavy timber) construction.

Framing		Solid-Sawn* (nominal)	Glulam* (actual)	SCL* (actual)
Floor	Columns	8×8	6¾ × 8¾	7 × 7½
	Beams	6×10	5×10½	5¼ × 9½
Roof	Columns	6×8	5 × 8¼	5¼ × 7½
	Beams**	4×6	3×6⅞	3½ × 5½

Framing		Minimum Thickness***
CLT	Floors	4"
	Roof	3"

* Minimum width by depth in inches. See IBC 2018, 2021 or 2024, Section 2304.11: Heavy timber construction, or IBC 2015, Section 602.4: Type IV.
** 3" nominal width allowed where sprinklered
*** Minimum thickness in inches. See IBC 2018, 2021, or 2024, Section 2304.11.3.1: Cross-laminated timber floors and Section 2304.11.4.1: Cross-laminated timber roofs

Fire Resistance



The National Design Standard for Wood Construction, an IBC-referenced standard

MINIMUM SIZE REQUIREMENTS FOR FIRE-RATED CLT

Alongside the 2015 IBC inclusion of CLT, the 2015 National Design Specification (NDS) for Wood Construction was expanded to include the calculated fire resistance of exposed CLT based on char depth and residual section capacity. CLT lamination thickness impacts calculations, as thinner layers char more quickly.

Establishing the construction type, and therefore, the required FRR, is the responsibility of the architect. Calculation of the FRR for structural elements is the responsibility of the structural engineer of record.

NDS Chapter 16, Fire Design of Wood Members, includes calculations and design procedures for exposed wood members. As with other structural materials, up to a 2-hour rating is allowable for CLT. As of 2025, there are no NDS provisions for calculating a 3-hour rating for CLT or other members. When required, this must be achieved through encapsulation with non-combustible materials or full-scale fire testing through alternative means and methods.

Effective Char Depths, a_{char} (in.)
(for CLT with $\beta_n = 1.5\text{in./hr}$)

Required Fire Endurance	Lamination Thicknesses, h_{lam} (in.)								
	$\frac{5}{8}$	$\frac{3}{4}$	$\frac{7}{8}$	1	$1\frac{1}{4}$	$1\frac{3}{8}$	$1\frac{1}{2}$	$1\frac{3}{4}$	2
1 Hour	2.2	2.2	2.1	2.0	2.0	1.9	1.8	1.8	1.8
1½ Hour	3.4	3.2	3.1	3.0	2.9	2.8	2.8	2.8	2.6
2 Hour	4.4	4.3	4.1	4.0	3.9	3.8	3.6	3.6	3.6

Fire Resistance

FIRE TESTING

Full-scale fire tests play a critical role in advancing the understanding of mass timber fire performance and developing new code provisions that expand the allowable use of mass timber. Tests commonly evaluate charring, structural performance under load, connection behavior, and fire containment under realistic conditions.

To support the introduction of new construction types for tall wood in the 2021 IBC, numerous full-scale fire tests were conducted to validate fire resistance calculations. As a result, jurisdictions adopting the 2021 IBC (or later) may follow prescriptive code approval paths for tall mass timber construction.

In some instances, a performance-based approval approach may be required. This includes projects in jurisdictions that have not yet adopted the 2021 IBC (or later), projects that propose assemblies or conditions not addressed explicitly by code provisions, or projects that exceed prescriptive limits. In these scenarios, fire testing may be used to demonstrate compliance under alternative means and methods.



Fire tests from the National Bureau of Alcohol, Tobacco, and Fire Arms (ATF) outside Washington D.C.

STRUCTURAL MODULE 1

Structural Basics of Mass Timber

Gravity Design of CLT

STRUCTURAL LIMIT STATES

A limit state is the condition beyond which a structure or component no longer fulfills its intended function. Structural engineers often evaluate horizontally spanning members such as CLT panels in terms of "limit states."

- Example: A panel that is loaded until it reaches the maximum allowable flexural stress (or maximum allowed moment) has reached a flexure limit state.

For CLT under a given load, consider what would be the span that reaches:

- The flexure limit state? (maximum bending stress)
- The shear limit state?
- The short-term deflection limit state? (mostly live load)
- The long-term deflection limit state? (live load plus dead load including creep)
- The vibration performance limit state?

The minimum span among all of the above limit states is the governing limit state. This sets the maximum span under the given load and boundary conditions.



Wood Innovation and Design Centre, Prince George, BC, Canada, 2014;
Architect: Michael Green Architecture; Structural Engineer: Equilibrium Consulting

STRUCTURAL MODULE 1

Structural Basics of Mass Timber

Gravity Design of CLT

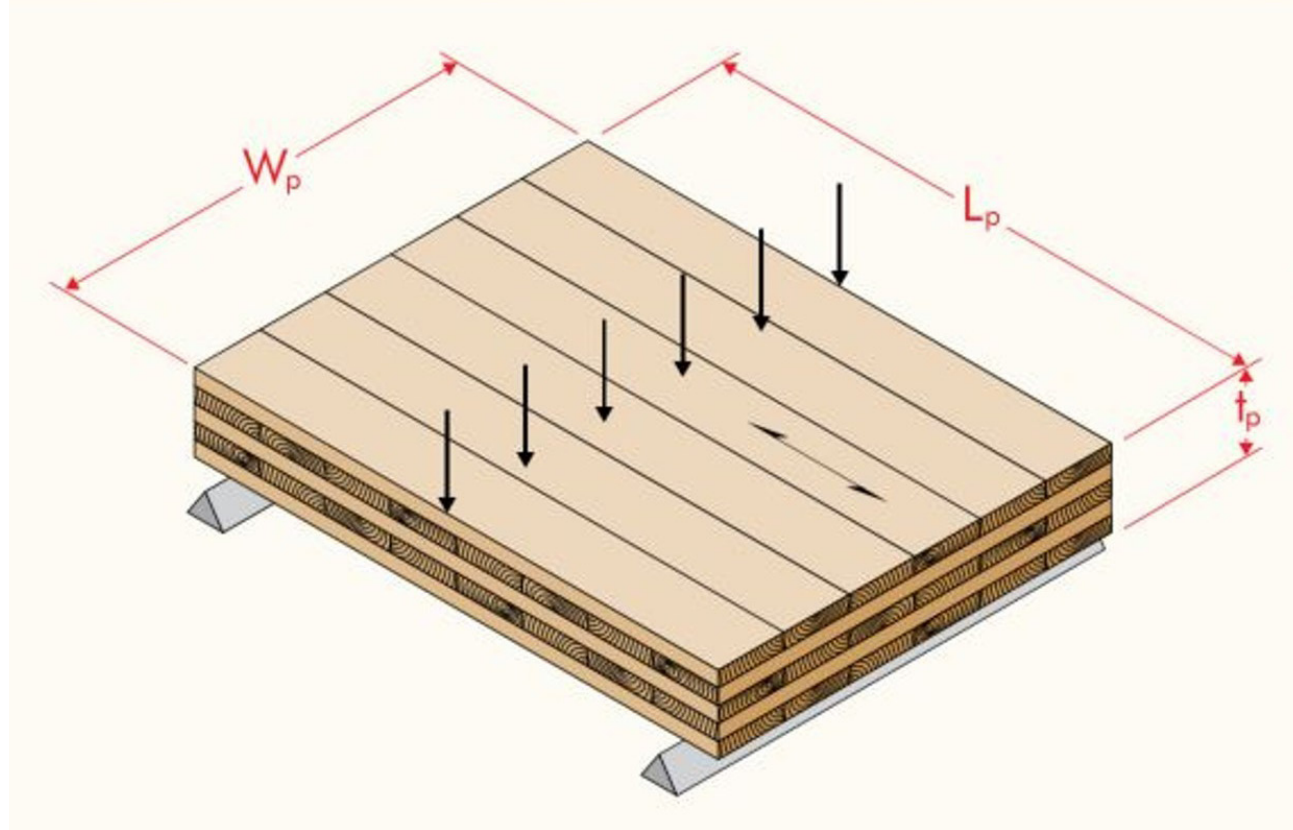


Diagram of CLT flatwise bending in the major strength direction

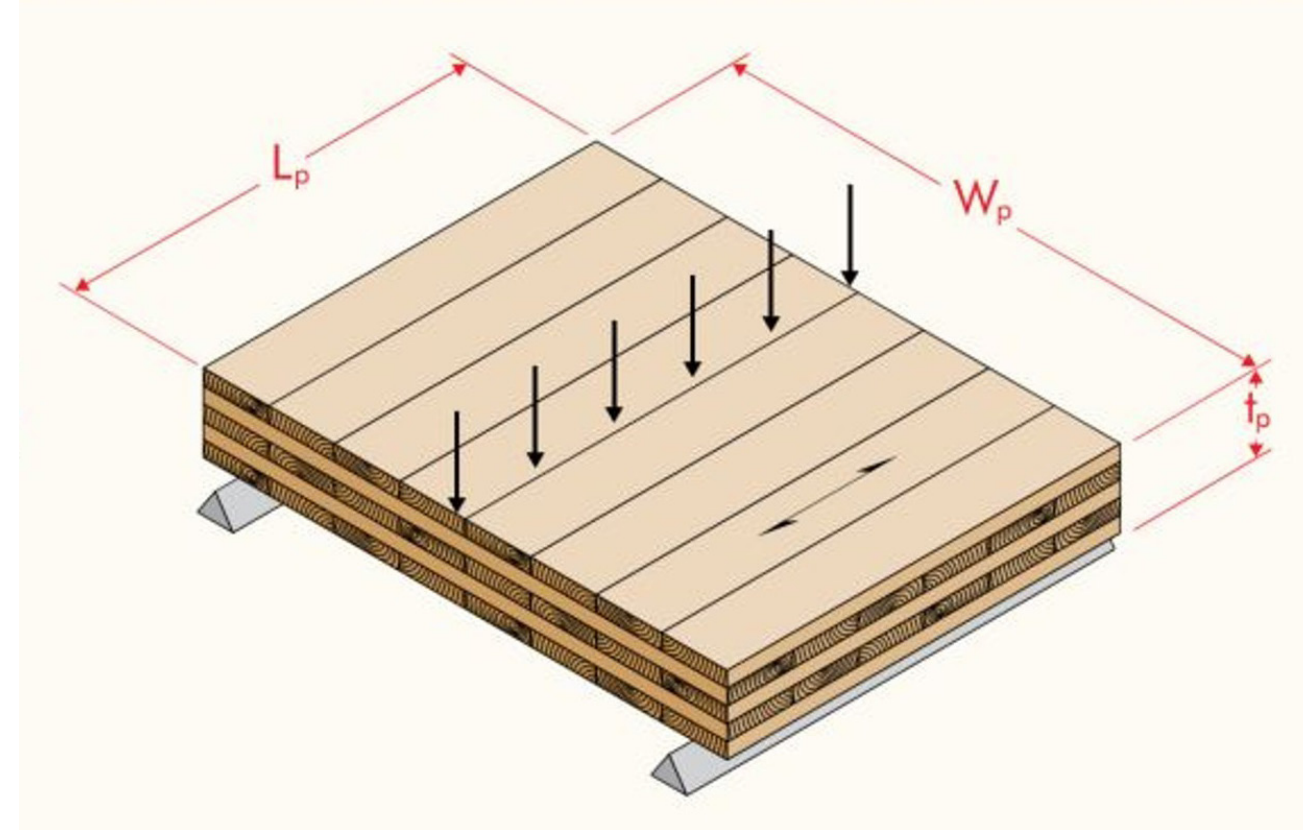


Diagram of CLT flatwise bending in the minor strength direction

STRONG AXIS VS. WEAK AXIS OF CLT

CLT is orthotropic: its structural properties differ along perpendicular directions. CLT panels are generally laid up in odd numbers of layers, creating a strong axis and a weak axis.

The strong axis is oriented in the direction of the top and bottom plies. This is the spanning direction, where fibers are subjected to tension and compression flexural stresses parallel to grain.

In the weak axis, top and bottom plies are oriented perpendicular to the spanning direction, subjecting fibers to tension and compression flexural stresses perpendicular to the grain.

Gravity Design of CLT

FLEXURAL STRENGTH

CLT behaves differently from homogeneous materials because its "weak" and "strong" layers carry stresses differently, contributing to unequal bending resistance. To simplify the design process, manufacturers publish effective section properties that account for the different layers, but still allow engineers to use conventional flexural calculations for homogeneous materials.*

$$M_b \leq C_D C_M C_t C_L (F_b S_{eff})$$

Per NDS

Typically = 1

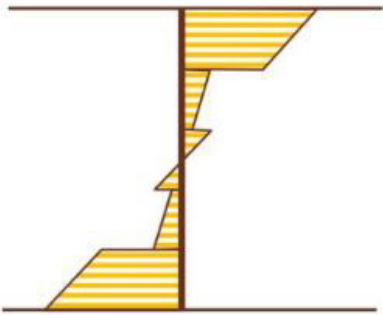
Provided as combined value by manufacturer

$$M_b \leq C_D (F_b S_{eff})$$

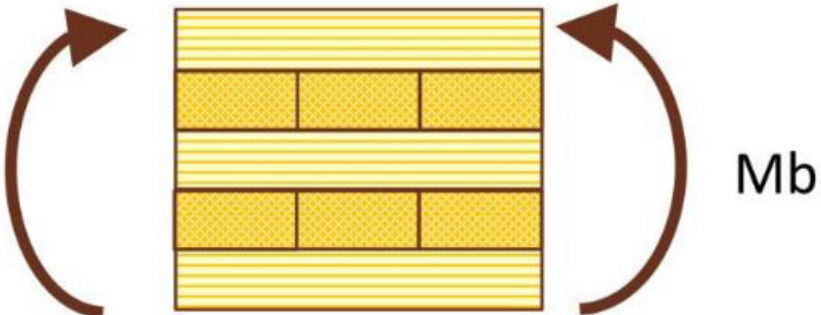
* Allowable bending moment M_b is calculated as the product of the allowable stress F_b (a function of species and grade) and the section modulus S (a function of the member dimensions). For CLT, the product of $F_b S$ is published as a single number for each panel type, and the allowable moment is further modified by four factors: C_D (duration), C_M (moisture content), C_t (temperature), and C_L (lateral stability).

PRG 320 ASD REFERENCE DESIGN VALUES: FLEXURE

The PRG 320 manufacturing standard includes tables of flexural strength for both V- and E-rated standard CLT layups (3-, 5-, and 7-ply) in both the major and minor strength directions. Because manufacturers can meet the standard using different species, grades, lamination configurations, and dimensions, each manufacturer publishes product-specific tables of structural properties and limit states.



Bending stress distribution across longitudinal and transverse plies



Typical bending in the strong axis of CLT, with tension on the bottom, compression on the top

ASD REFERENCE DESIGN VALUES^a FOR BASIC CLT GRADES AND LAYUPS (FOR USE IN THE U.S.)

CLT Grade	t _p (in.)	Lamination Thickness (in.) in CLT Layup								Major Strength Direction				Minor Strength Direction			
		=		⊥		=		⊥		(F _b S) _{eff,0} (lbf-ft/ft of width)	(EI) _{eff,0} (10 ⁶ lbf-in. ² /ft of width)	(GA) _{eff,0} (10 ⁶ lbf/ft of width)	V _{s,0} (lbf/ft of width)	(F _b S) _{eff,90} (lbf-ft/ft of width)	(EI) _{eff,90} (10 ⁶ lbf-in. ² /ft of width)	(GA) _{eff,90} (10 ⁶ lbf/ft of width)	V _{s,90} (lbf/ft of width)
E1	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					4,525	115	0.46	1,490	160	3.1	0.61	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			10,400	440	0.92	2,480	1,370	81	1.2	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,375	1,089	1.4	3,475	3,150	313	1.8	2,480
E2	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					3,825	102	0.53	1,980	165	3.6	0.56	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			8,825	389	1.1	3,300	1,440	95	1.1	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,600	963	1.6	4,625	3,300	364	1.7	3,300
E3	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					2,800	81	0.35	1,160	110	2.3	0.44	385
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			6,400	311	0.69	1,930	955	61	0.87	1,160
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	11,325	769	1.0	2,700	2,210	234	1.3	1,930
E4	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					4,525	115	0.50	1,820	140	3.4	0.62	605
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			10,400	440	1.0	3,025	1,230	88	1.2	1,820
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,400	1,089	1.5	4,225	2,850	338	1.9	3,025
E5	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					3,825	101	0.46	1,650	160	3.1	0.55	550
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			8,800	389	0.92	2,750	1,370	81	1.1	1,650
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,575	962	1.4	3,850	3,150	312	1.7	2,750
V1	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					2,090	108	0.53	1,980	165	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			4,800	415	1.1	3,300	1,440	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,500	1,027	1.6	4,625	3,300	364	1.8	3,300
V1(N)	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					1,980	108	0.53	1,980	150	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			4,550	415	1.1	3,300	1,300	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,025	1,027	1.6	4,625	3,000	364	1.8	3,300
V2	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					2,030	95	0.46	1,490	160	3.1	0.52	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			4,675	363	0.91	2,480	1,370	81	1.0	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,275	898	1.4	3,475	3,150	312	1.6	2,480

Gravity Design of CLT

ROLLING SHEAR

The development of CLT introduced a new term to the structural engineering lexicon: rolling shear.

Rolling shear is named for the mechanism of transferring shear perpendicular to the wood grain, through the cell structure of wood as if the metaphorical "straws in a bundle" are rolling over one another. This behavior is not seen in traditional solid-sawn or glulam structures, where wood fibers are primarily aligned with the direction of stress.

In CLT, rolling shear limits the bending resistance of panels. It is particularly critical in applications that rely on two-way action, such as point-supported floor systems.

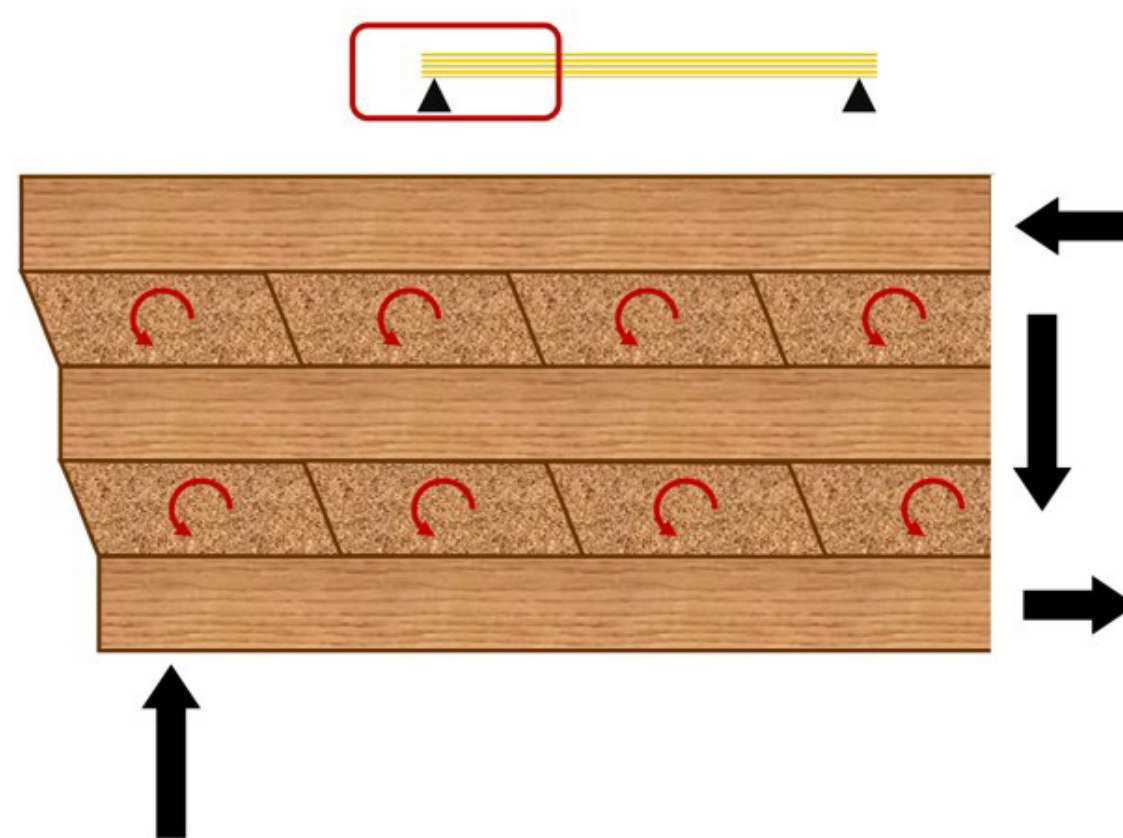


Diagram of rolling shear, as seen in CLT

PLANAR SHEAR	=	OUT-OF-PLANE SHEAR	=	ROLLING SHEAR
Wood Structural Panel Term		Structural Engineering Term		CLT Term

Comparable shear terminology

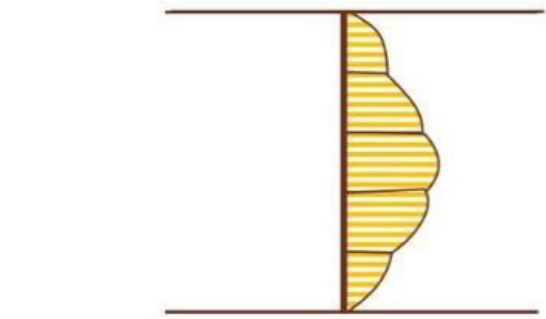
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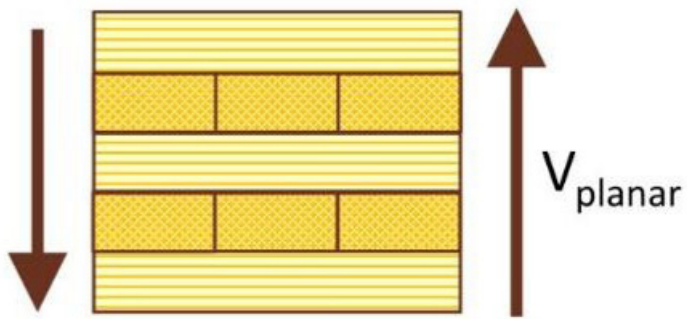
Gravity Design of CLT

SHEAR STRENGTH

Shear behavior in CLT is also influenced by its layered construction, with "weak" and "strong" layers contributing differently to shear resistance. As with flexure, the shear limit state is calculated using effective properties that account for the different properties of the layers using equivalent section properties.



Shear stress distribution across longitudinal and transverse plies



Typical shear in the strong axis of CLT, with transverse plies experiencing rolling shear

$$V_{\text{planar}} \leq C_M C_t (F_s (I_b Q)_{\text{eff}})$$

Typically = 1 Provided as combined value by manufacturer

$$V_{\text{planar}} \leq V_s$$

PRG 320 ASD REFERENCE DESIGN VALUES: SHEAR

PRG 320 includes reference shear strength values for standard CLT layups in both principal directions. As with flexural properties, manufacturers publish shear design values for their specific products, which should be used for project-specific design calculations.

ASD REFERENCE DESIGN VALUES^a FOR BASIC CLT GRADES AND LAYUPS (FOR USE IN THE U.S.)

CLT Grade	t _p (in.)	Lamination Thickness (in.) in CLT Layup								Major Strength Direction				Minor Strength Direction			
		=	⊥	=	⊥	=	⊥	=	⊥	(F _s) _{eff,f,0} (lb _f -ft/ft of width)	(EI) _{eff,f,0} (10 ⁶ lb _f -in. ² /ft of width)	(GA) _{eff,f,0} (10 ⁶ lb _f /ft of width)	V _{s,0} (lb _f /ft of width)	(F _s) _{eff,f,90} (lb _f -ft/ft of width)	(EI) _{eff,f,90} (10 ⁶ lb _f -in. ² /ft of width)	(GA) _{eff,f,90} (10 ⁶ lb _f /ft of width)	V _{s,90} (lb _f /ft of width)
E1	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	4,525	115	0.46	1,490	160	3.1	0.61	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	10,400	440	0.92	2,480	1,370	81	1.2	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,375	1,089	1.4	3,475	3,150	313	1.8	2,480
E2	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	3,825	102	0.53	1,980	165	3.6	0.56	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,825	389	1.1	3,300	1,440	95	1.1	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,600	963	1.6	4,625	3,300	364	1.7	3,300
E3	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2,800	81	0.35	1,160	110	2.3	0.44	385
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	6,400	311	0.69	1,930	955	61	0.87	1,160
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	11,325	769	1.0	2,700	2,210	234	1.3	1,930
E4	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	4,525	115	0.50	1,820	140	3.4	0.62	605
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	10,400	440	1.0	3,025	1,230	88	1.2	1,820
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,400	1,089	1.5	4,225	2,850	338	1.9	3,025
E5	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	3,825	101	0.46	1,650	160	3.1	0.55	550
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,800	389	0.92	2,750	1,370	81	1.1	1,650
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,575	962	1.4	3,850	3,150	312	1.7	2,750
V1	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2,090	108	0.53	1,980	165	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	4,800	415	1.1	3,300	1,440	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,500	1,027	1.6	4,625	3,300	364	1.8	3,300
V1(N)	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1,980	108	0.53	1,980	150	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	4,550	415	1.1	3,300	1,300	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,025	1,027	1.6	4,625	3,000	364	1.8	3,300
V2	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	2,030	95	0.46	1,490	160	3.1	0.52	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	4,675	363	0.91	2,480	1,370	81	1.0	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,275	898	1.4	3,475	3,150	312	1.6	2,480

Gravity Design of CLT

FLEXURE AND SHEAR DEFORMATION

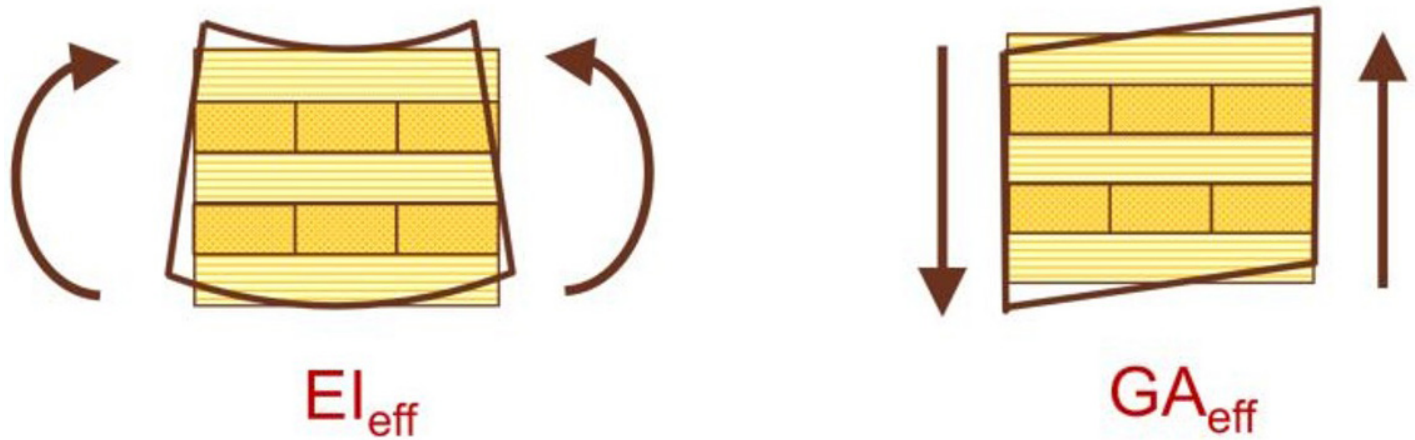
Deformations and deflections are also calculated using effective section properties for bending (EI_{eff}) and shear (GA_{eff}). The effective properties of the whole section are based on the material (elastic moduli) and dimensional thickness of each ply.

$$(EI)_{eff,f,0} = \sum_{i=2}^{n-1} E_i b_i \frac{t_i^3}{12} + \sum E_i b_i t_i z_i^2$$

$$(GA)_{eff,f,0} = \frac{(t_p - \frac{t_1}{2} - \frac{t_n}{2})^2}{[(\frac{t_1}{2G_i b_0}) + (\sum_{i=2}^{n-1} \frac{t_i}{G_i b_0}) + (\frac{t_n}{2G_n b_0})]}$$

PRG 320 ASD REFERENCE DESIGN VALUES: DEFORMATION

PRG 320 provides deformation-related properties for common CLT layups, allowing engineers to evaluate expected deflections under design loads. The deformation properties shown are for short-term loading. Deformations under long-term sustained loads will increase due to the creep behavior of wood.



Effective properties and deformation modes under flexure (left) and shear (right)

ASD REFERENCE DESIGN VALUES^a FOR BASIC CLT GRADES AND LAYUPS (FOR USE IN THE U.S.)

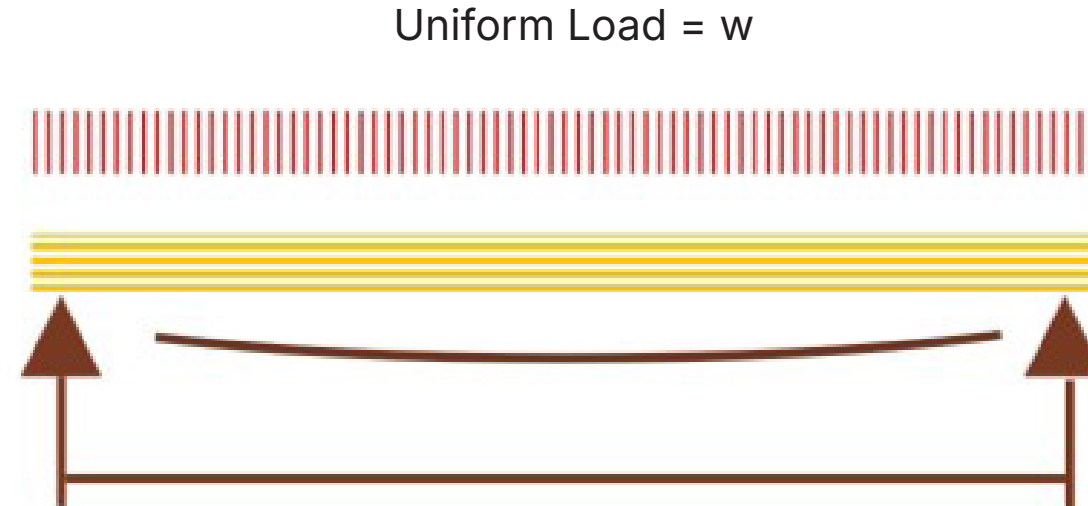
CLT Grade	t _p (in.)	Lamination Thickness (in.) in CLT Layup								Major Strength Direction				Minor Strength Direction			
		=	⊥	=	⊥	=	⊥	=		(F _t S) _{eff,f,0} (lbf-ft/ft of width)	(EI) _{eff,f,0} (10 ⁶ lbf-in. ² /ft of width)	(GA) _{eff,f,0} (10 ⁶ lbf/ft of width)	V _{s,0} (lbf/ft of width)	(F _t S) _{eff,f,90} (lbf-ft/ft of width)	(EI) _{eff,f,90} (10 ⁶ lbf-in. ² /ft of width)	(GA) _{eff,f,90} (10 ⁶ lbf/ft of width)	V _{s,90} (lbf/ft of width)
E1	4 1/8	1 3/8	1 3/8	1 3/8						4,525	115	0.46	1,490	160	3.1	0.61	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				10,400	440	0.92	2,480	1,370	81	1.2	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,375	1,089	1.4	3,475	3,150	313	1.8	2,480
E2	4 1/8	1 3/8	1 3/8	1 3/8						3,825	102	0.53	1,980	165	3.6	0.56	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				8,825	389	1.1	3,300	1,440	95	1.1	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,600	963	1.6	4,625	3,300	364	1.7	3,300
E3	4 1/8	1 3/8	1 3/8	1 3/8						2,800	81	0.35	1,160	110	2.3	0.44	385
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				6,400	311	0.69	1,930	955	61	0.87	1,160
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	11,325	769	1.0	2,700	2,210	234	1.3	1,930
E4	4 1/8	1 3/8	1 3/8	1 3/8						4,525	115	0.50	1,820	140	3.4	0.62	605
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				10,400	440	1.0	3,025	1,230	88	1.2	1,820
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,400	1,089	1.5	4,225	2,850	338	1.9	3,025
E5	4 1/8	1 3/8	1 3/8	1 3/8						3,825	101	0.46	1,650	160	3.1	0.55	550
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				8,800	389	0.92	2,750	1,370	81	1.1	1,650
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,575	962	1.4	3,850	3,150	312	1.7	2,750
V1	4 1/8	1 3/8	1 3/8	1 3/8						2,090	108	0.53	1,980	165	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				4,800	415	1.1	3,300	1,440	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,500	1,027	1.6	4,625	3,300	364	1.8	3,300
V1(N)	4 1/8	1 3/8	1 3/8	1 3/8						1,980	108	0.53	1,980	150	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				4,550	415	1.1	3,300	1,300	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,025	1,027	1.6	4,625	3,000	364	1.8	3,300
V2	4 1/8	1 3/8	1 3/8	1 3/8						2,030	95	0.46	1,490	160	3.1	0.52	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8				4,675	363	0.91	2,480	1,370	81	1.0	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,275	898	1.4	3,475	3,150	312	1.6	2,480

Gravity Design of CLT

FLEXURE AND SHEAR DEFORMATION

Deflection in CLT panels is evaluated for both short- and long-term loading conditions. Short-term deflection is primarily associated with live loads, while long-term deflection includes dead loads and creep.

Creep accounts for wood deforming gradually over time under a sustained load. Long-term deflection of a CLT panel can be significantly greater than instantaneous deflection.



Short-term mid-span deflection of a uniformly loaded one-way slab

$$\Delta_{ST} = \frac{5}{384} \times \frac{wL^4}{EI^{eff}} + \frac{1}{8} \times \frac{wL^2}{GA^{eff}}$$

Total long-term mid-span deflection of a uniformly loaded one-way slab

$$\Delta_T = K_{cr} \Delta_{LT} + \Delta_{ST}$$

Δ_{ST} = Deflection due to short term loading

Δ_{LT} = Immediate deflection due to long term loading

K_{cr} = Creep factor for CLT = 2.0 for dry service conditions

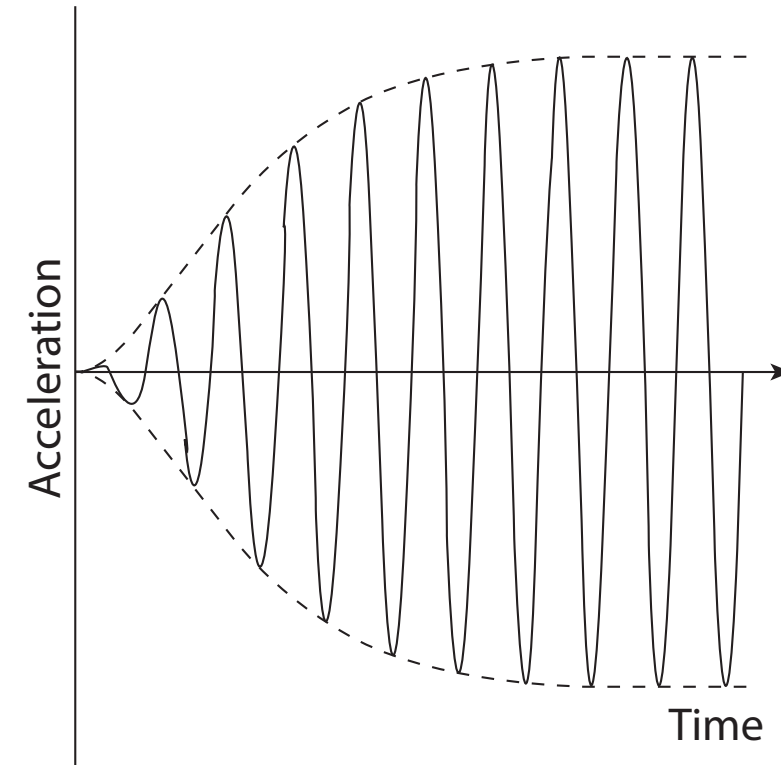
Vibration

Vibration is difficult to quantify because it is ultimately a subjective measure of personal comfort. Acceptable vibration levels vary by architectural program, with higher levels of sensitivity for residences and office spaces, as well as the type of excitation, such as walking, running, or vibrating machinery.

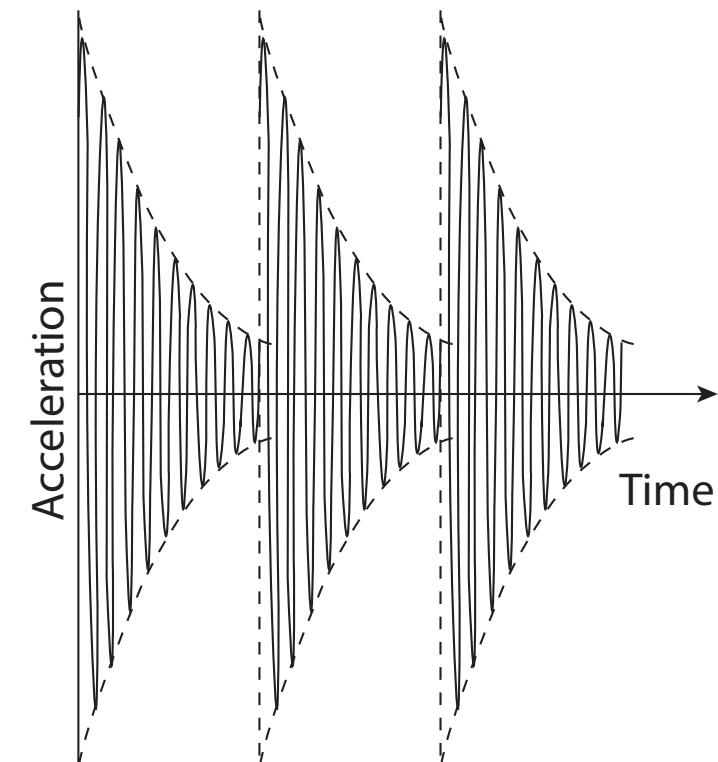
Floor vibration performance is governed by both natural frequency and acceleration response. The natural frequency affects how quickly a floor vibrates when disturbed. Lower frequencies are commonly perceived as more flexible or "bouncy."

Acceleration response describes the intensity of motion felt by occupants under dynamic loading, such as walking. People are particularly sensitive to acceleration response: even small accelerations can result in discomfort.

As with acoustical performance, the lightness of CLT can pose challenges for vibration control. The high strength and low mass of CLT make it more susceptible to perceptible levels of vibration than concrete. As a result, for most mass timber floor systems, vibration is the governing limit state.

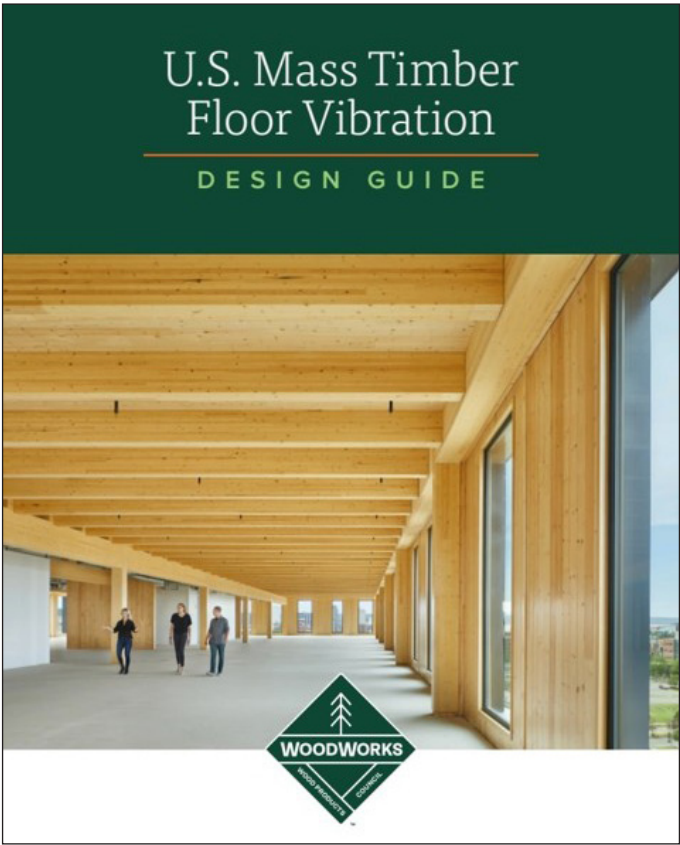


Acceleration build-up to resonant response of damped system

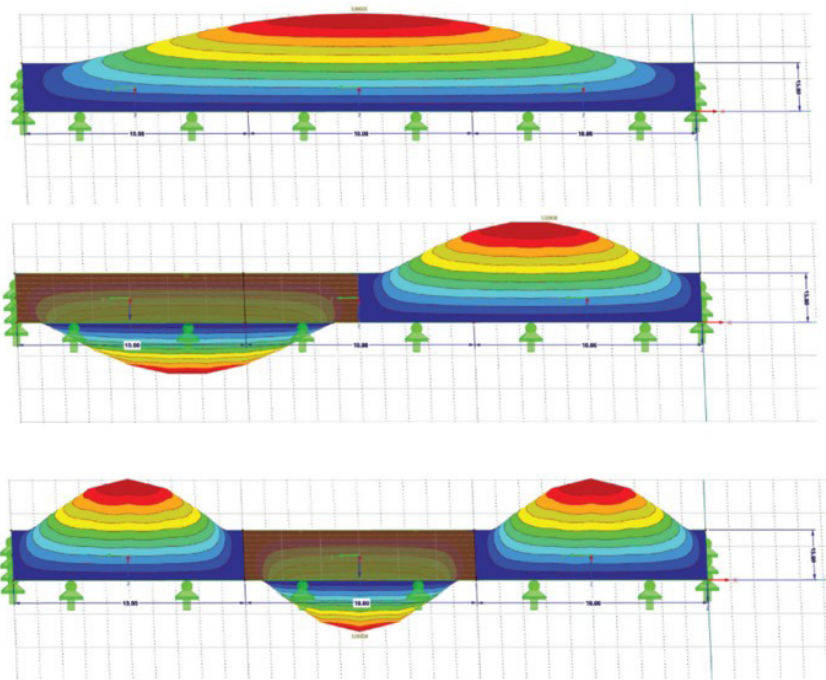


Acceleration build-up to transient response of damped system

Vibration



Free mass timber floor vibration guide, published by WoodWorks



Deformed shapes of an edge-supported panel, representing the first three natural modes of vibration. Analogous to a plucked guitar string

VIBRATION ANALYSIS

The choice of analysis method reflects a tradeoff between effort and precision, and is often informed by building program, span length, and the structural complexity of the floor system.

Empirical Calculations

Simplified empirical calculations are commonly used during early design stages to evaluate simple-span CLT panels. The equations are based on limited assumptions and do not reflect the full complexity of an individual building, but remain useful for preliminary assessment.

The U.S. Mass Timber Floor Vibration Design Guide by WoodWorks is the most comprehensive reference for vibration analysis and is widely used in U.S. practice. This free, downloadable resource presents equations for preliminary evaluation of natural frequency and acceleration response for common CLT floor configurations.

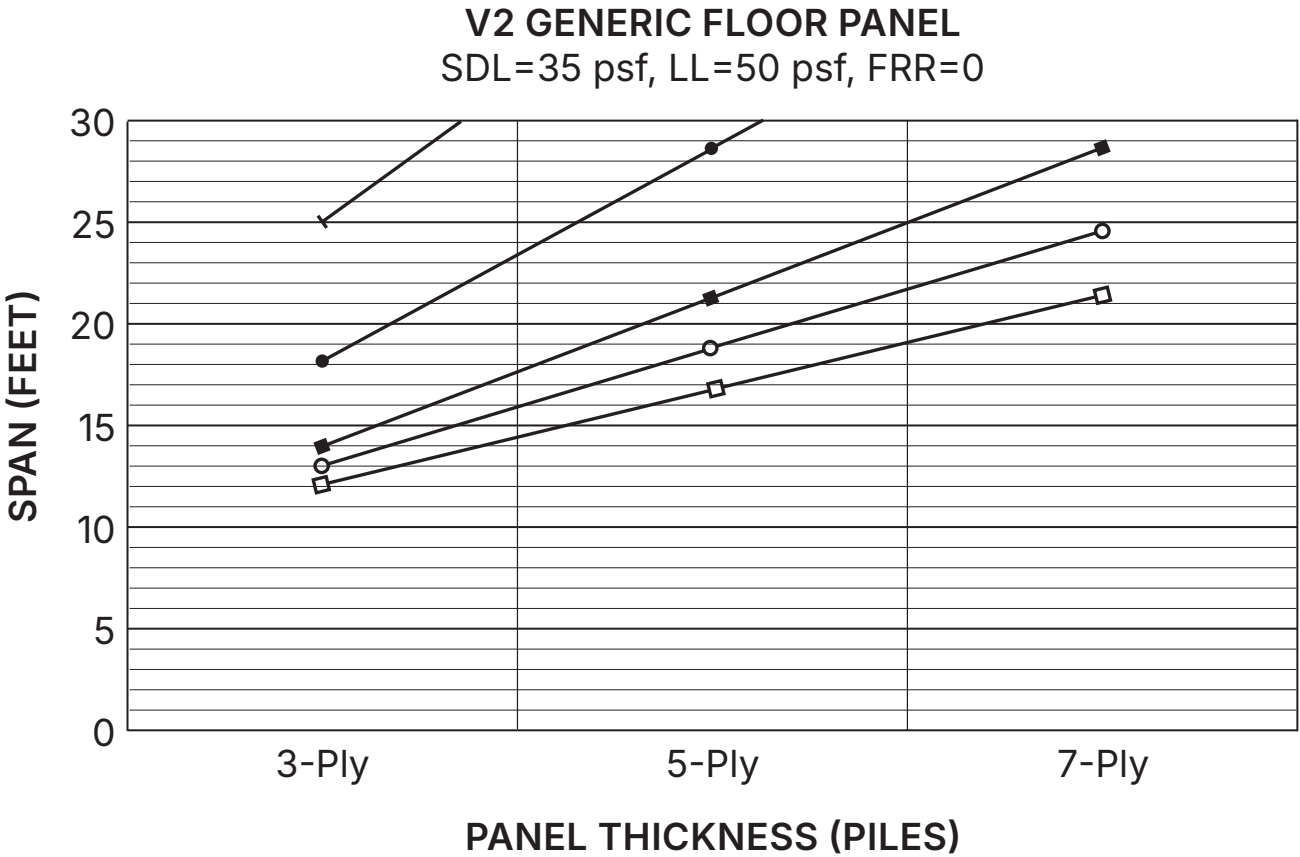
The equation below limits allowable span using a simplified equation for simple-span CLT panels on rigid supports.

$$\text{Maximum Vibration-} \\ \text{Controlled Span } L = \frac{1}{12.05} \times \frac{(EI_{app})^{0.293}}{(\rho A)^{0.122}}$$

Finite Element Analysis (FEA)

Finite element analysis is a more advanced vibration analysis method using actual boundary conditions, panel layups, and the properties of each layer of CLT. Though FEA is significantly more complex, expensive, and time consuming, it can more accurately predict vibration performance, facilitating more efficiently designed floor systems.

Vibration



GOVERNING LIMIT STATES FOR UNRATED CLT FLOORS

The graph at the left compares maximum allowable spans for unrated (0-hour FRR) V2 2-span continuous CLT floor panels under typical office loading for each limit state: flexure, shear, total load deflection, live load deflection, and vibration.

The lowest span for each panel thickness is the governing limit for the maximum allowable span. Since the line for vibration is the lowest, vibration is the governing limit state across 3-ply, 5-ply, and 7-ply thicknesses, with spans of 12', 17', and 21.5' respectively.

- Shear
- Live Load Deflection
- Total Load Deflection
- Flexure
- Vibration

TYPICAL SPAN RANGES FOR UNRATED CLT

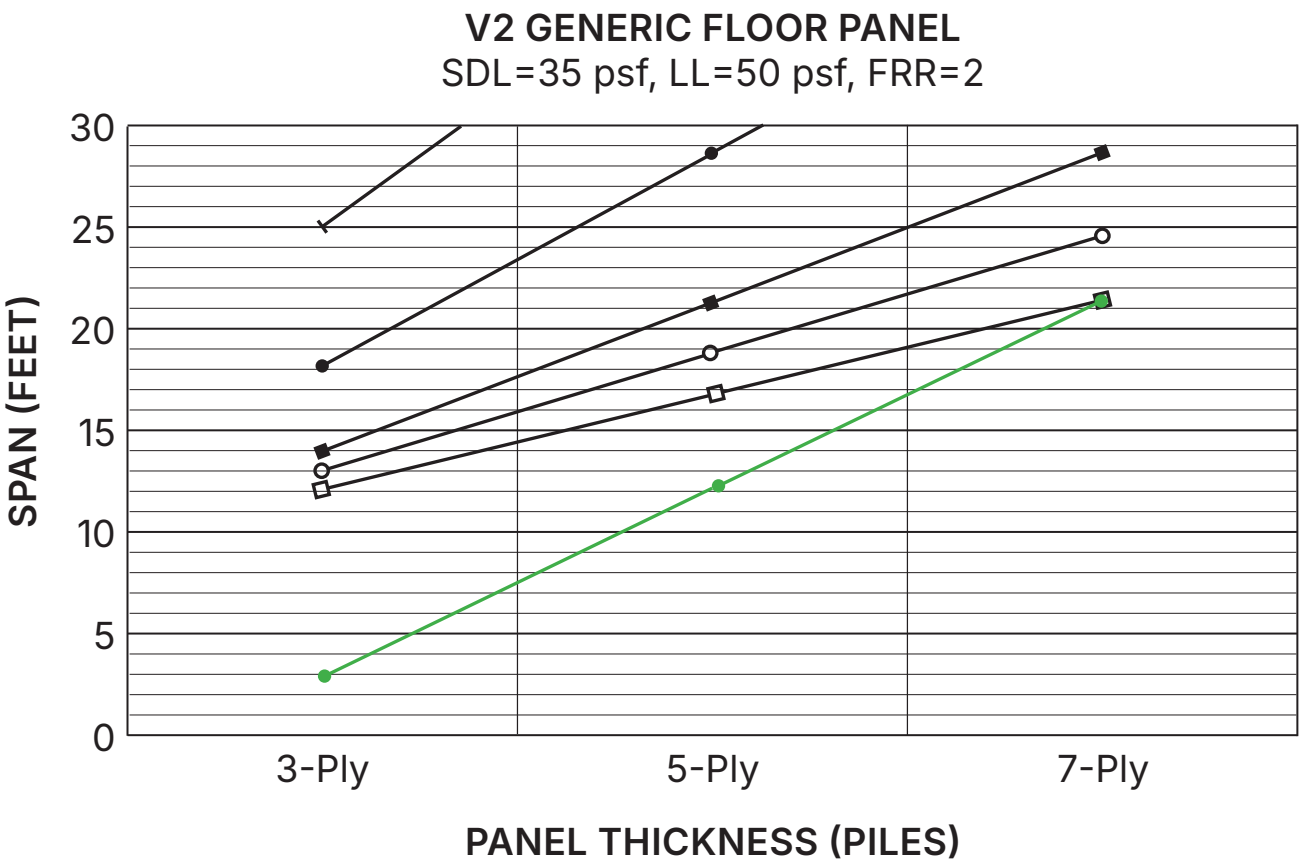
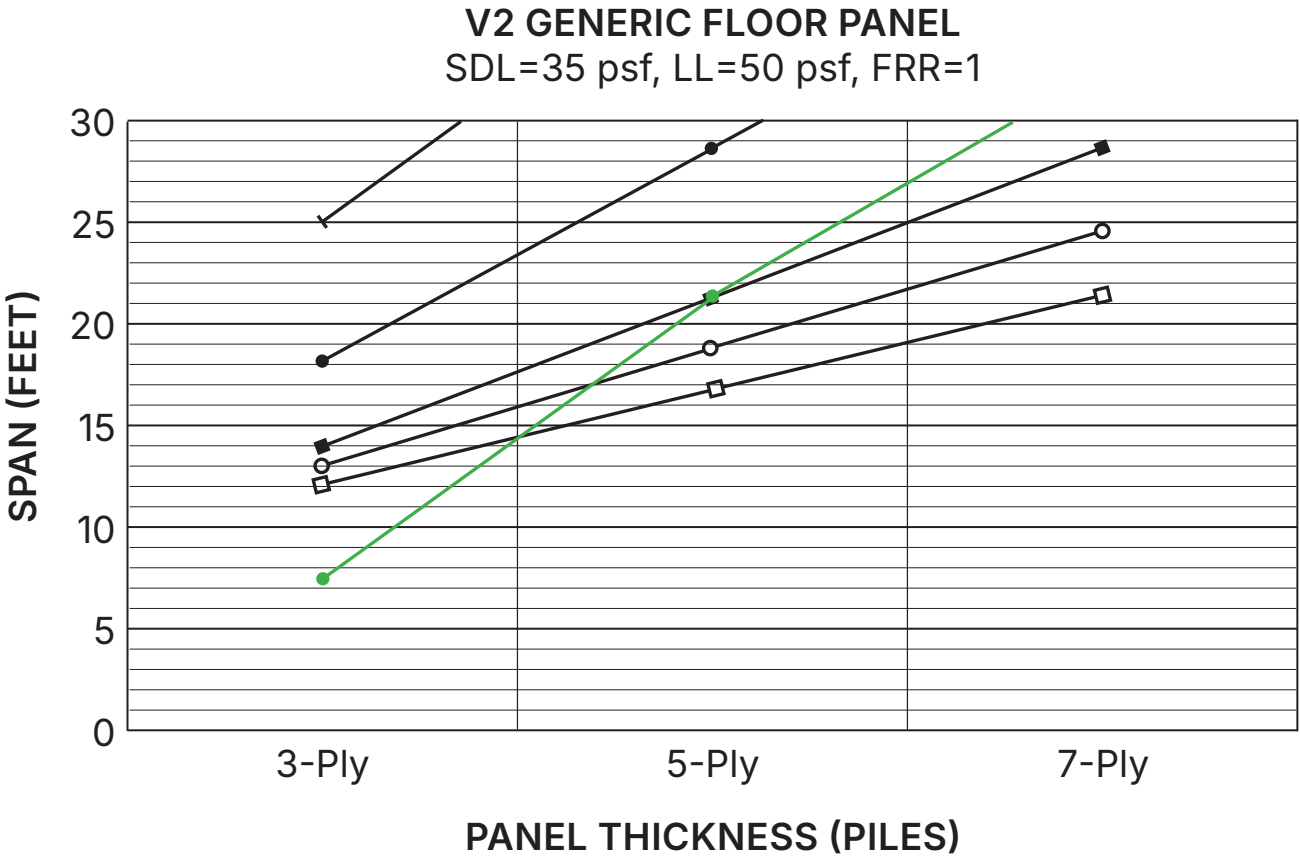
Calculating the spanning capabilities of CLT in different conditions is complex and dependent on project-specific criteria.

However, typical span ranges give a reasonable starting point for architects in the conceptual design phase. It is also useful to have rules of thumb for reasonable spans in common, simple conditions:

- Floor spans are almost always governed by vibration criteria
- Roof spans are almost always governed by flexural strength

TYPICAL SPAN RANGES FOR UNRATED CLT		
CLT	Floor (DL + 50 psf)	Roof (DL + 30 psf)
3-Ply	11' - 12'	12' - 14'
5-Ply	16' - 17'	18' - 20'
7-Ply	20' - 21'	24' - 25'

Vibration



GOVERNING LIMIT STATES FOR FIRE-RATED CLT FLOORS

The graphs at the left compare the maximum allowable spans for 1-hour and 2-hour rated V2 2-span continuous CLT floor panels under typical office loading for each limit state. Introducing fire-resistance ratings adds an additional limit state, post-char flexure. This is the limit state under a special load combination after a fire has burned through the char layer.

For 1-hour fire rated floors, post-char flexure is the governing limit state for 3-ply CLT, but the vibration limit state continues to govern for 5-ply and 7-ply panels. 3-ply panels are uncommon in fire rated construction, so the 1-hour FRR does not have a significant impact on spans.

For 2-hour fire rated construction, post-char flexure is the governing limit state for 3-ply, 5-ply, and 7-ply CLT. 3-ply is effectively unusable for 2-hour FRR, given its limited 7.5' span; the span of 5-ply is reduced by approximately 1/3, to 12'; and the maximum allowable span of 7-ply remains unchanged at 21.5'.

- Shear
- Live Load Deflection
- Total Load Deflection
- Flexure
- Vibration
- Post-Char Flexure

Vibration

DESIGN DECISIONS

Vibration mitigation is not the sole responsibility of structural engineers. Architects play a direct and consequential role in shaping vibration performance through early design decisions, including:

- Structural grid layout
- Span length
- Panel thickness and continuity
- Floor toppings and composite action
- Location of vibration-sensitive occupancies

Because vibration is often the governing limit state for CLT floors, early coordination of architectural intent and structural strategy is essential to achieving vibration performance goals, occupant comfort, and cost efficiency.



Vibration control is critical for craftsmanship at a watchmaking facility, so a concrete topping was added to the Omega Factory's CLT floor deck. Omega Factory, Biel/Bienne, Switzerland, 2017; Architect: Shigeru Ban Architects, Structural Engineer: Création Holz, SJB Kempter Fitze, Schnetzer Puskas Ingenieure

STRUCTURAL MODULE 1

Structural Basics of Mass Timber

Connections

Mass timber connections are a primary design consideration, influencing both the structure and appearance of a project. Though connection design may seem less visually prominent than species selection, material choices, or architectural form, connections can have a strong visual impact.

Connections play a critical role in structural and fire performance and can be costly. Connection design should be considered as early as possible, as it often influences member sizing, structural grids, and overall layouts.

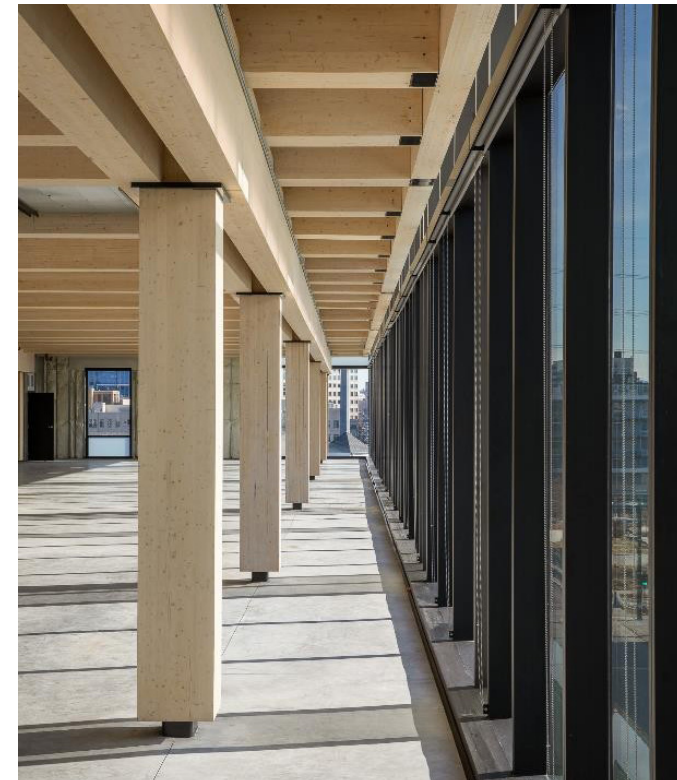
Design considerations include:

- Aesthetics
- Strength
- Constructibility
- Construction tolerance
- Shrinkage
- Moisture protection
- Fire resistance
- Cost

Many manufacturers offer connection design as part of their pre-construction services, leveraging their expertise to use their products and systems efficiently.



First Tech Federal Credit Union Office, Hillsboro, OR, United States, 2018; Architect: Hacker Architects, Structural Engineer: Kramer Gehlen



Platte Fifteen, Denver, CO, United States, 2019; Architect: OZ Architecture, Structural Engineer: KL&A Engineers & Builders

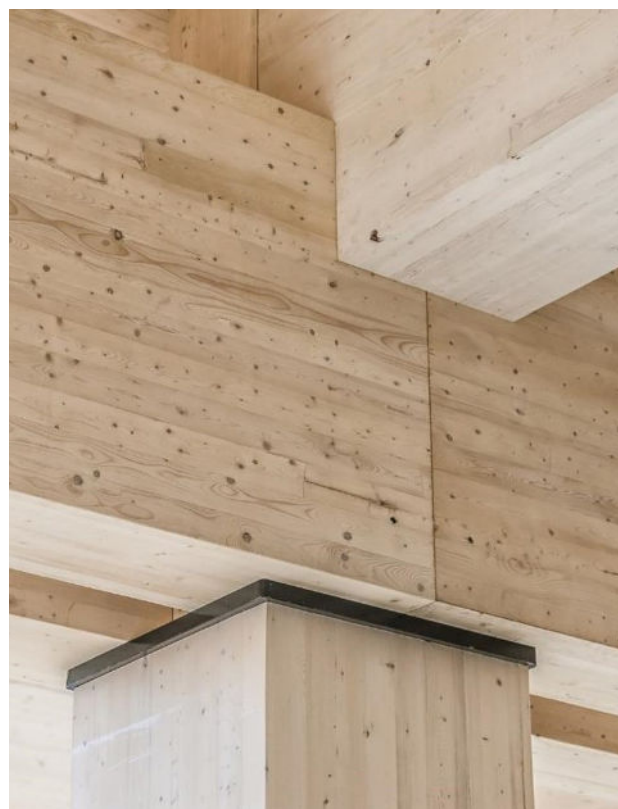
STRUCTURAL MODULE 1

Structural Basics of Mass Timber

Connections



Mass timber structure with both bearing and concealed proprietary connections



Bearing connection



Assembly diagram of concealed proprietary connection system

CONNECTION TYPES

Mass timber connections can be broadly categorized based on how forces are transferred and whether the system is standardized or bespoke.

- **Bearing Connections**
Simple connections that transfer loads primarily through compression between wood members, or between wood and another material. Typically the least expensive and simplest to fabricate and install
- **Fastener-Based Connections**
Loads are transferred through discrete dowel-type fasteners like bolts, pins, or screws. These connections may offer greater design flexibility, but often introduce higher localized stresses perpendicular to the grain

The categories can be further classified into two subcategories, which indicate how the connections are designed and procured:

- **Proprietary Connections**
Pre-engineered and tested proprietary systems with documented structural and fire performance characteristics. Often concealed rather than exposed. Proprietary connections can simplify detailing, but may limit design flexibility and tie a project to a specific manufacturer
- **Custom Connections**
Designed for specific structural, architectural, or constructibility requirements of a project. Offers greater flexibility, but requires additional engineering, coordination, and cost

**STRUCTURAL
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Connections

CONNECTION CONSIDERATIONS

- **Aesthetics**
Connection detailing may enhance or detract from architectural expression. Exposed connections such as hangers, plates, or fasteners may inadvertently become focal points in a space, while concealed connections create a clean, uninterrupted appearance
- **Load Direction and Grain Orientation**
Connections are often designed to align primary stresses parallel to grain, where wood has the greatest strength and stiffness. This limits long-term deformations and relative movement between timber members
- **Shrinkage**
Steel components that pass through CLT panels are often designed to bypass perpendicular-to-grain compression, reducing panel shrinkage and minimizing deformation



Economical exposed beam hanger



Concealed connections



Continuous column-to-column connections avoid perpendicular to grain stresses at beams and floor plates



Steel column base recessed in CLT floor

**STRUCTURAL
MODULE 1**

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Connections



Unprotected steel connections are only allowed in unrated construction

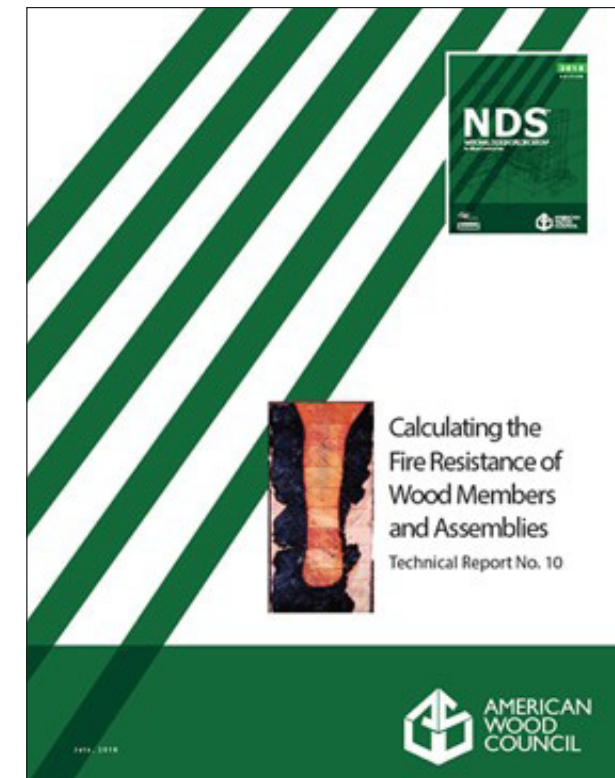


Fire testing of timber member with concealed steel connections

FIRE RATED CONNECTIONS

Fire ratings are required by code for specific occupancy types, building heights, and construction types. Some mass timber buildings may include unrated assemblies where code allows, typically in smaller, low-occupancy buildings.

In rated construction, connections must also meet the fire rating requirement. A common strategy is to install steel fasteners within the char layer, so they are protected from direct fire exposure.



Technical report with empirical methods for determining fire ratings for timber members and connections

**STRUCTURAL
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Basics of
Mass Timber**

**MASS
TIMBER
PEDAGOGY**

Connections

PANEL-TO-PANEL CONNECTIONS

Mass timber panels are prefabricated as large structural members, but buildings require panels to be joined together to form floors, walls, and roofs. Panel-to-panel connections transfer loads, maintain alignment, and control movement between panels. This connection detailing impacts structural integrity, fire performance, and constructibility.

Half-Lap Connections

A half-lap connection reduces the thickness of each panel along its edge so the panels overlap flush. This traditional joint is seen historically in heavy timber construction. The main drawback of half-lap connections is that they require a predetermined construction sequence.

Spline Connections

A spline connection joins panels using a thin strip of wood inserted into a groove at the edge of a panel. This common connection type allows for slight adjustments in alignment and is widely used in contemporary mass timber assemblies.

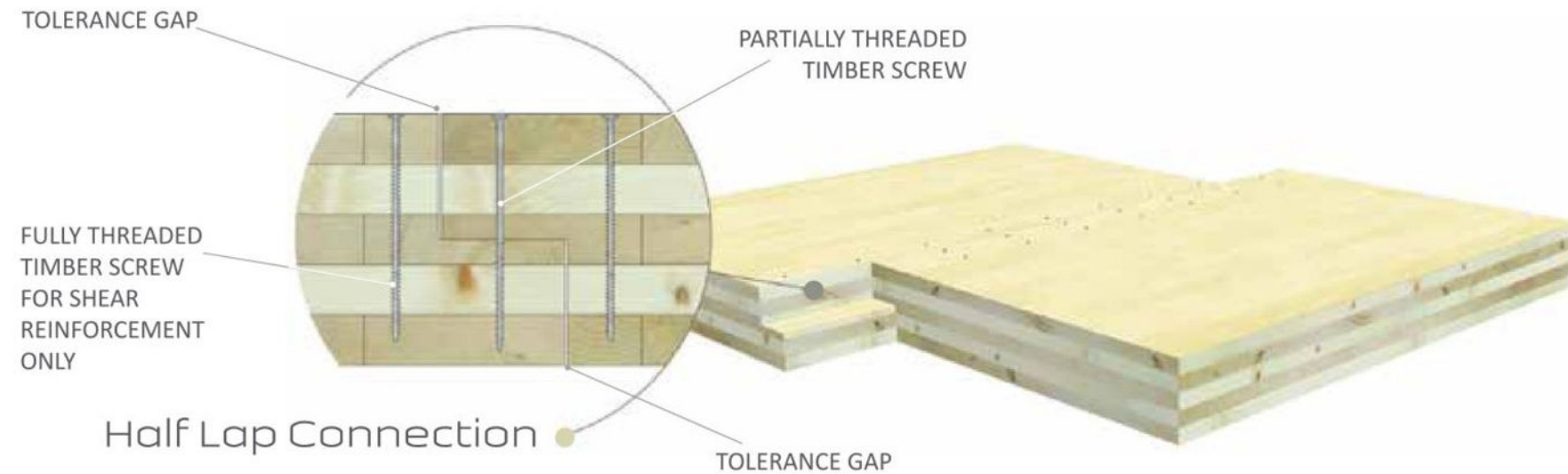


Diagram of a half-lap connection between two CLT panels

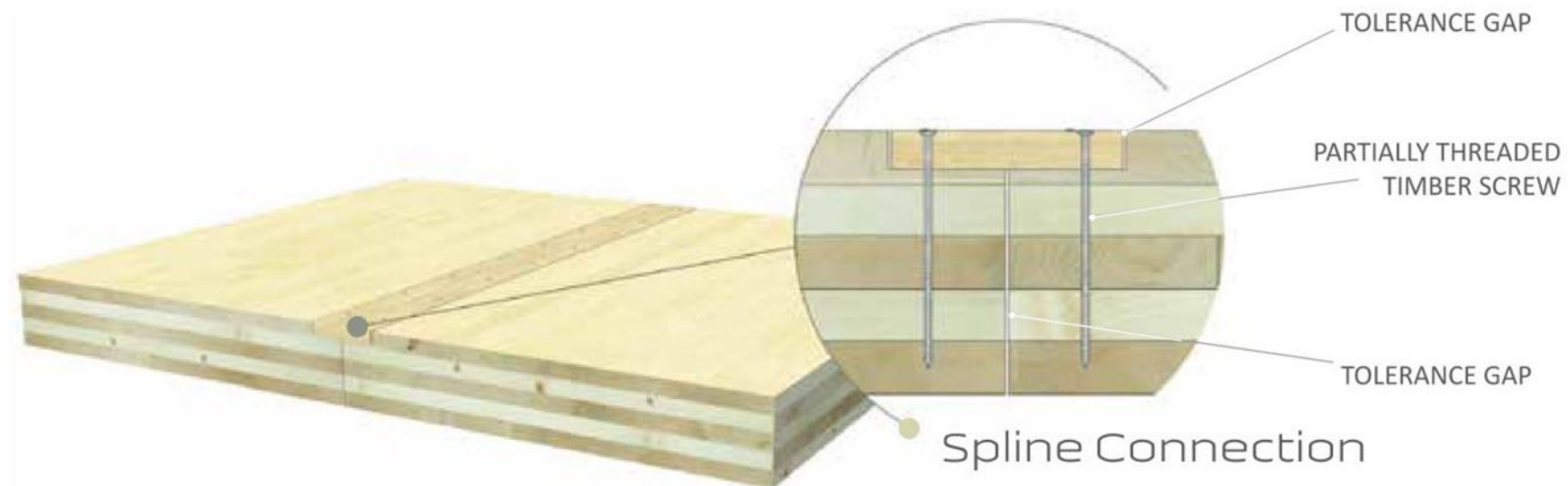


Diagram of a spline connection between two CLT panels

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Conclusion

A foundational understanding of the structural basics of mass timber engineering begins with an understanding of wood as a material. Wood is not monolithic: its structural performance varies within forests, among species, and between softwoods and hardwoods. Its organic qualities, such as anisotropic behavior and sensitivity to moisture, distinguish wood from concrete and steel.

Building on this material foundation, mass timber introduces a range of engineered wood products, each with distinct performance characteristics and implications pertaining to aesthetics, cost, availability, and environmental responsibility. For example, the project-specific benefits of CLT versus MPP, or glulam versus LVL, must be considered in relation to code requirements, fire resistance, vibration, member sizing, spans, and connection detailing.

Together, these principles form the technical basis of mass timber structural design.



World of Volvo, Gothenburg, Sweden, 2024; Architect; Henning Larsen, Structural Engineer: BRA Teknik

STRUCTURAL MODULE 1

Structural Basics of Mass Timber

Citations

IMAGES, TABLES, AND GRAPHS

Page 01

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MODULE 1

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